

Line Segments

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Coordinate Geometry (ref cg2)

This document contains all 70 *Line Segments* questions, each with a fully worked solution and a TikZ diagram on the coordinate plane: endpoints, segments, midpoints, points of division, triangles, quadrilaterals and circles are all plotted. Several *find the unknown coordinate* questions are shown as *locus* diagrams — a fixed point with a line and a circle whose intersections are the solutions.

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Line Segments 01 (cg2-001)

Difficulty: Foundation / *Marks: 4*

Find both (i) the midpoint and (ii) the exact length of the segment from $(-2, 4)$ to $(4, 12)$.

Worked solution.

Step 1: (i) Midpoint:

$$M = \left(\frac{-2 + 4}{2}, \frac{4 + 12}{2} \right) = (1, 8)$$

Using the midpoint formula: average the x-coordinates and average the y-coordinates separately.

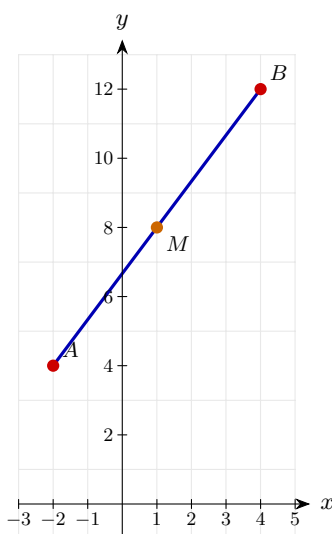
Step 2: (ii) Length:

$$d = \sqrt{(4 - (-2))^2 + (12 - 4)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$$

The distance formula comes from Pythagoras applied to the horizontal and vertical differences. Note that $4 - (-2) = 6$, not 2 — be careful with double negatives.

Final answer: (i) $M = (1, 8)$; (ii) $d = 10$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 02 (cg2-002)

Difficulty: Foundation / Marks: 4

Find both (i) the midpoint and (ii) the exact length of the segment from $(3, -5)$ to $(-3, 3)$.

Worked solution.

Step 1: (i) Midpoint:

$$M = \left(\frac{3 + (-3)}{2}, \frac{-5 + 3}{2} \right) = (0, -1)$$

Using the midpoint formula: average the x-coordinates and average the y-coordinates separately.

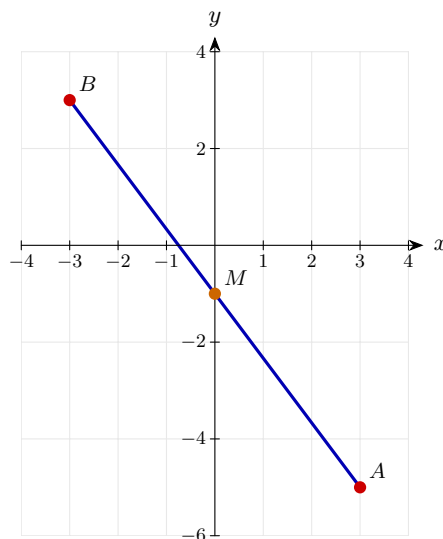
Step 2: (ii) Length:

$$d = \sqrt{(-3 - 3)^2 + (3 - (-5))^2} = \sqrt{36 + 64} = \sqrt{100} = 10$$

Using the distance formula. Note that $3 - (-5) = 8$, not -2 . Subtracting a negative is the same as adding.

Final answer: (i) $M = (0, -1)$; (ii) $d = 10$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 03 (cg2-003)

Difficulty: Foundation / Marks: 4

Find the exact length of the segment of the line $y = 3x + 1$ between $x = 2$ and $x = 5$.

Worked solution.

Step 1: Find the y-coordinates at each x-value.

$$x = 2 : y = 3(2) + 1 = 7 \quad x = 5 : y = 3(5) + 1 = 16$$

Substitute each x-value into the line equation to find the corresponding y-coordinate, giving the two endpoints.

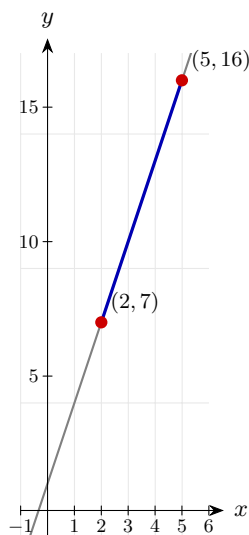
Step 2: Apply the distance formula between $(2, 7)$ and $(5, 16)$.

$$d = \sqrt{(5 - 2)^2 + (16 - 7)^2} = \sqrt{9 + 81} = \sqrt{90} = 3\sqrt{10}$$

$$\sqrt{90} = \sqrt{9 \times 10} = 3\sqrt{10}.$$

Final answer: $d = 3\sqrt{10}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 04 (cg2-004)

Difficulty: Foundation / *Marks: 4*

Find the exact length of the segment of the line $y = -2x + 7$ between $x = 1$ and $x = 4$.

Worked solution.

Step 1: Find the y-coordinates.

$$x = 1 : y = -2(1) + 7 = 5 \quad x = 4 : y = -2(4) + 7 = -1$$

Substitute each x-value into the equation to find the two endpoints of the segment.

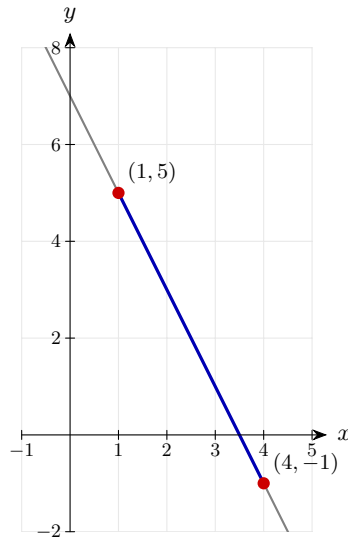
Step 2: Apply the distance formula between $(1, 5)$ and $(4, -1)$.

$$d = \sqrt{(4 - 1)^2 + (-1 - 5)^2} = \sqrt{9 + 36} = \sqrt{45} = 3\sqrt{5}$$

$$\sqrt{45} = \sqrt{9 \times 5} = 3\sqrt{5}.$$

Final answer: $d = 3\sqrt{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 05 (cg2-005)

Difficulty: Foundation / Marks: 5

Find the midpoint and exact length of the segment of $y = \frac{1}{2}x - 3$ between $x = 2$ and $x = 10$.

Worked solution.

Step 1: Find the endpoints.

$$x = 2 : y = \frac{1}{2}(2) - 3 = -2 \quad x = 10 : y = \frac{1}{2}(10) - 3 = 2$$

Points are $A(2, -2)$ and $B(10, 2)$.

Step 2: Midpoint:

$$M = \left(\frac{2 + 10}{2}, \frac{-2 + 2}{2} \right) = (6, 0)$$

Using the midpoint formula. The y-coordinates cancel ($-2 + 2 = 0$), so the midpoint lies on the x-axis.

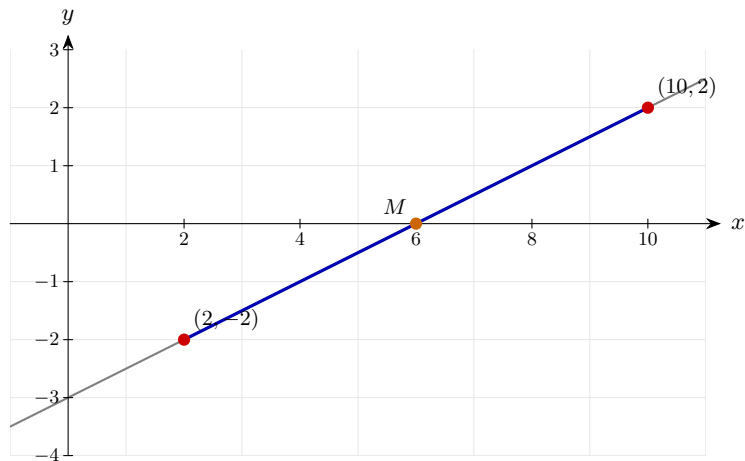
Step 3: Length:

$$d = \sqrt{(10 - 2)^2 + (2 - (-2))^2} = \sqrt{64 + 16} = \sqrt{80} = 4\sqrt{5}$$

$$\sqrt{80} = \sqrt{16 \times 5} = 4\sqrt{5}.$$

Final answer: $M = (6, 0); d = 4\sqrt{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 06 (cg2-006)

Difficulty: Foundation / Marks: 4

Find the exact length of the segment of the line $2x + y - 4 = 0$ between $x = 0$ and $x = 3$.

Worked solution.

Step 1: Rearrange the line to $y = -2x + 4$, then find the endpoints.

$$x = 0 : y = 4 \quad x = 3 : y = -2(3) + 4 = -2$$

Rearrange the equation to make y the subject first, then substitute each x -value to find the endpoints.

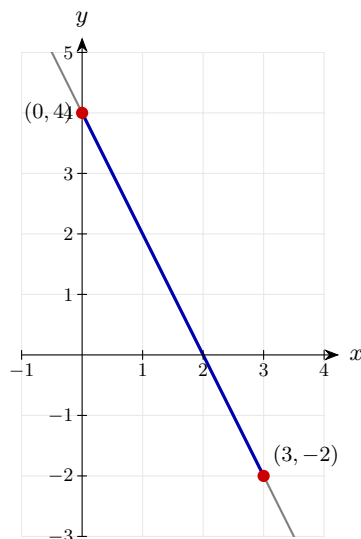
Step 2: Apply the distance formula between $(0, 4)$ and $(3, -2)$.

$$d = \sqrt{(3 - 0)^2 + (-2 - 4)^2} = \sqrt{9 + 36} = \sqrt{45} = 3\sqrt{5}$$

Simplify the surd: find the largest perfect square factor of 45, which is 9.

Final answer: $d = 3\sqrt{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 07 (cg2-007)

Difficulty: Foundation / Marks: 5

Find the midpoint and exact length of the segment of $x + 2y - 8 = 0$ between $x = 2$ and $x = 6$.

Worked solution.

Step 1: Rearrange: $y = \frac{8-x}{2}$. Find endpoints.

$$x = 2 : y = \frac{6}{2} = 3 \quad x = 6 : y = \frac{2}{2} = 1$$

Points are $A(2, 3)$ and $B(6, 1)$.

Step 2: Midpoint:

$$M = \left(\frac{2+6}{2}, \frac{3+1}{2} \right) = (4, 2)$$

Using the midpoint formula: average the x-coordinates and average the y-coordinates.

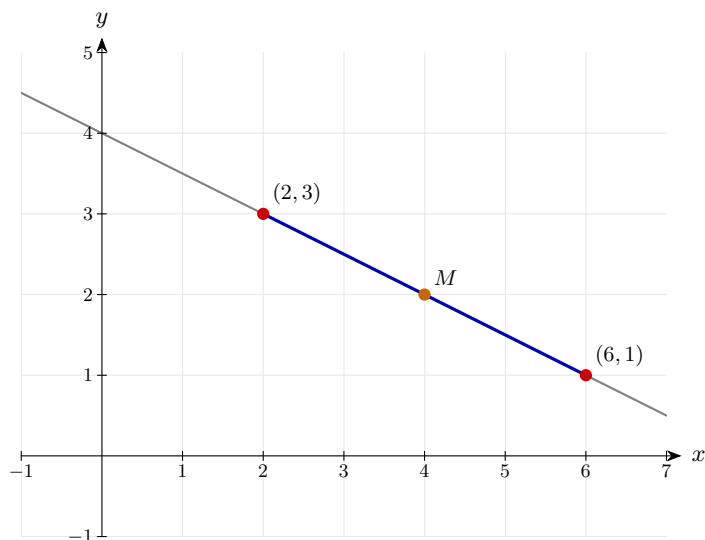
Step 3: Length:

$$d = \sqrt{(6-2)^2 + (1-3)^2} = \sqrt{16+4} = \sqrt{20} = 2\sqrt{5}$$

$$\sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}.$$

Final answer: $M = (4, 2); d = 2\sqrt{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 08 (cg2-008)

Difficulty: Foundation / Marks: 5

Find the midpoint and exact length of the segment of $y = 4x - 3$ between $x = -1$ and $x = 2$.

Worked solution.

Step 1: Find the endpoints.

$$x = -1 : y = 4(-1) - 3 = -7 \quad x = 2 : y = 4(2) - 3 = 5$$

Points are $A(-1, -7)$ and $B(2, 5)$.

Step 2: Midpoint:

$$M = \left(\frac{-1 + 2}{2}, \frac{-7 + 5}{2} \right) = (0.5, -1)$$

Using the midpoint formula. The x-coordinate gives $1/2 = 0.5$, which is fine to leave as a decimal or fraction.

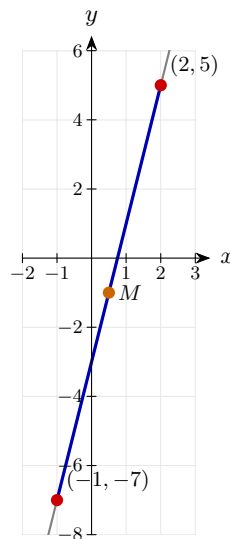
Step 3: Length:

$$d = \sqrt{(2 - (-1))^2 + (5 - (-7))^2} = \sqrt{9 + 144} = \sqrt{153} = 3\sqrt{17}$$

$$\sqrt{153} = \sqrt{9 \times 17} = 3\sqrt{17}.$$

Final answer: $M = (0.5, -1); d = 3\sqrt{17}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 09 (cg2-009)

Difficulty: Foundation / Marks: 3

The midpoint of segment AB is $M(3, 6)$. If $A = (1, 2)$, find the coordinates of B .

Worked solution.

Step 1: Use the midpoint formula in reverse: $x_B = 2 \times x_M - x_A$.

$$x_B = 2(3) - 1 = 5$$

Rearranging the midpoint formula: since the midpoint x-coordinate equals $(x_A + x_B)/2$, we get $x_B = 2 * x_M - x_A$.

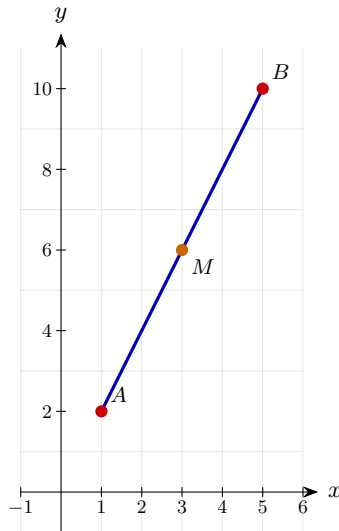
Step 2: Similarly for y:

$$y_B = 2(6) - 2 = 10$$

Same technique for the y-coordinate: $y_B = 2 * y_M - y_A$.

Final answer: $B = (5, 10)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 10 (cg2-010)

Difficulty: Foundation / Marks: 3

The midpoint of segment PQ is $M(-1, 4)$. If $P = (3, 2)$, find Q .

Worked solution.

Step 1: Find x_Q :

$$x_Q = 2(-1) - 3 = -5$$

Using the rearranged midpoint formula: $x_Q = 2 * x_M - x_P$. A common mistake is to forget the factor of 2.

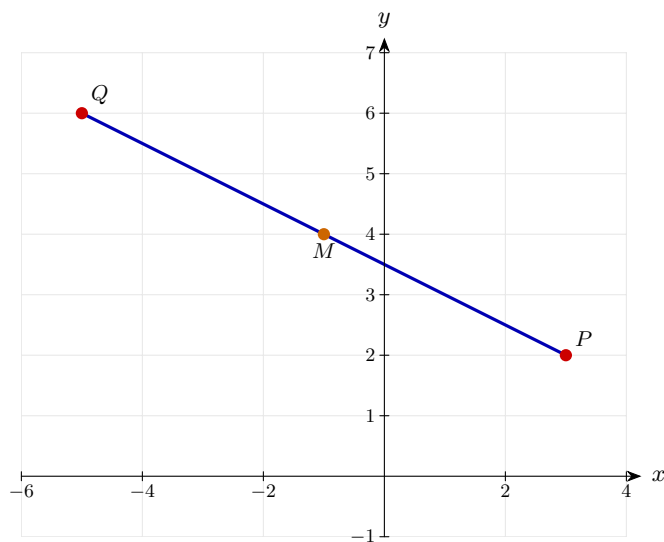
Step 2: Find y_Q :

$$y_Q = 2(4) - 2 = 6$$

Same approach for the y-coordinate: $y_Q = 2 * y_M - y_P$.

Final answer: $Q = (-5, 6)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 11 (cg2-011)

Difficulty: Foundation / Marks: 6

Points $A(a, 1)$ and $B(7, b)$ both lie on the line $y = x - 2$. Find (i) the values of a and b , (ii) the midpoint of AB , and (iii) the exact length of AB .

Worked solution.

Step 1: (i) Substitute $y = 1$ into $y = x - 2$ to find a .

$$1 = a - 2 \quad \Rightarrow \quad a = 3$$

Since $A(a, 1)$ lies on the line, its coordinates must satisfy the equation.

Step 2: Substitute $x = 7$ to find b .

$$b = 7 - 2 = 5$$

So $A = (3, 1)$ and $B = (7, 5)$.

Step 3: (ii) Midpoint:

$$M = \left(\frac{3+7}{2}, \frac{1+5}{2} \right) = (5, 3)$$

Using the midpoint formula with the coordinates found in parts (i). You can verify that $(5, 3)$ also lies on $y = x - 2$.

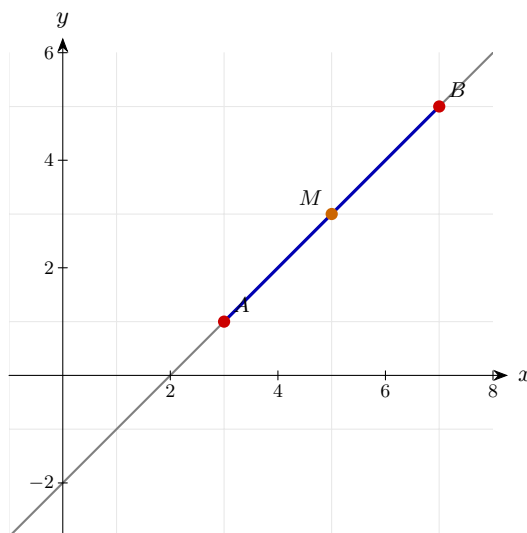
Step 4: (iii) Length:

$$d = \sqrt{(7-3)^2 + (5-1)^2} = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}$$

$$\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}.$$

Final answer: (i) $a = 3$, $b = 5$; (ii) $M = (5, 3)$; (iii) $d = 4\sqrt{2}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 12 (cg2-012)

Difficulty: Foundation / Marks: 6

Points $A(a, 1)$ and $B(b, 7)$ lie on the line $y = 3x - 5$. Find (i) the values of a and b , (ii) the midpoint of AB , (iii) the exact length of AB .

Worked solution.

Step 1: (i) Substitute $y = 1$ to find a :

$$1 = 3a - 5 \implies 3a = 6 \implies a = 2$$

Since $A(a, 1)$ lies on the line, substitute $y = 1$ and solve for x to find a .

Step 2: Substitute $y = 7$ to find b :

$$7 = 3b - 5 \implies 3b = 12 \implies b = 4$$

Points: $A = (2, 1)$, $B = (4, 7)$.

Step 3: (ii) Midpoint:

$$M = \left(\frac{2+4}{2}, \frac{1+7}{2} \right) = (3, 4)$$

Using the midpoint formula with the coordinates found above.

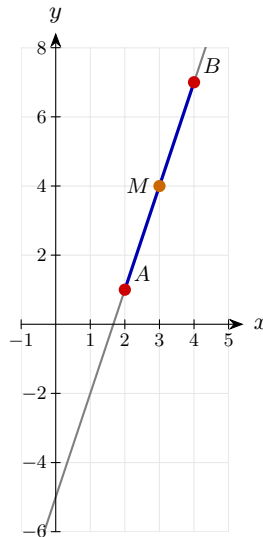
Step 4: (iii) Length:

$$d = \sqrt{(4-2)^2 + (7-1)^2} = \sqrt{4+36} = \sqrt{40} = 2\sqrt{10}$$

Simplify the surd: the largest perfect square factor of 40 is 4, so $\sqrt{40} = 2\sqrt{10}$.

Final answer: (i) $a = 2$, $b = 4$; (ii) $M = (3, 4)$; (iii) $d = 2\sqrt{10}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 13 (cg2-013)

Difficulty: Foundation / Marks: 6

Points $C(2, c)$ and $D(d, 1)$ lie on the line $y = 2x - 7$. Find (i) the values of c and d , (ii) the midpoint of CD , (iii) the exact length of CD .

Worked solution.

Step 1: (i) Find c : substitute $x = 2$ into $y = 2x - 7$.

$$c = 2(2) - 7 = 4 - 7 = -3$$

Since $C(2, c)$ lies on the line, substitute $x = 2$ directly into $y = 2x - 7$ to find the y -coordinate.

Step 2: Find d : substitute $y = 1$ into $y = 2x - 7$.

$$1 = 2d - 7 \implies 2d = 8 \implies d = 4$$

Here we know $y = 1$, so substitute into the equation and solve for x . This is the reverse of the previous step.

Step 3: (ii) Midpoint of $C(2, -3)$ and $D(4, 1)$.

$$M = \left(\frac{2+4}{2}, \frac{-3+1}{2} \right) = (3, -1)$$

Using the midpoint formula with the coordinates found in part (i).

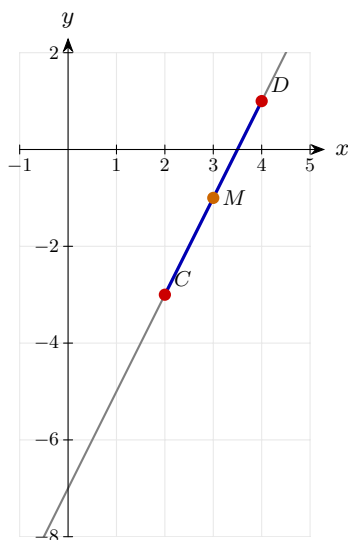
Step 4: (iii) Length of CD .

$$CD = \sqrt{(4-2)^2 + (1-(-3))^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$$

Note that $1 - (-3) = 4$. Simplify the surd: $\sqrt{20} = \sqrt{4 \cdot 5} = 2\sqrt{5}$.

Final answer: (i) $c = -3$, $d = 4$; (ii) $M = (3, -1)$; (iii) $2\sqrt{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 14 (cg2-014)

Difficulty: Foundation / Marks: 7

Points $P(p, 5)$ and $Q(7, q)$ lie on the line $3x - 2y - 1 = 0$. Find (i) p and q , (ii) the midpoint of PQ , (iii) the exact length of PQ .

Worked solution.

Step 1: (i) Substitute $y = 5$ to find p :

$$3p - 10 - 1 = 0 \implies 3p = 11 \implies p = \frac{11}{3}$$

Substitute $y = 5$ into the equation $3x - 2y - 1 = 0$ and solve for x . Fractional coordinates are perfectly valid.

Step 2: Substitute $x = 7$ to find q :

$$21 - 2q - 1 = 0 \implies 2q = 20 \implies q = 10$$

Points: $P = \left(\frac{11}{3}, 5\right)$, $Q = (7, 10)$.

Step 3: (ii) Midpoint:

$$M = \left(\frac{\frac{11}{3} + 7}{2}, \frac{5 + 10}{2} \right) = \left(\frac{\frac{32}{3}}{2}, \frac{15}{2} \right) = \left(\frac{16}{3}, \frac{15}{2} \right)$$

When averaging fractions, convert to a common denominator first: $\frac{11}{3} + 7 = \frac{11}{3} + \frac{21}{3} = \frac{32}{3}$, then divide by 2.

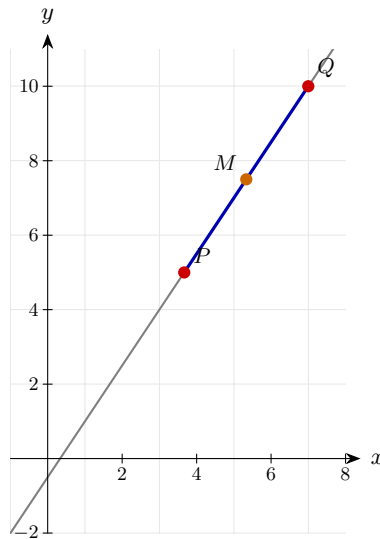
Step 4: (iii) Length:

$$d = \sqrt{\left(7 - \frac{11}{3}\right)^2 + (10 - 5)^2} = \sqrt{\left(\frac{10}{3}\right)^2 + 25} = \sqrt{\frac{100}{9} + \frac{225}{9}} = \sqrt{\frac{325}{9}} = \frac{5\sqrt{13}}{3}$$

Convert 25 to ninths ($\frac{225}{9}$) so both terms share a denominator. Then $\sqrt{325} = \sqrt{25 \cdot 13} = 5\sqrt{13}$.

Final answer: (i) $p = \frac{11}{3}$, $q = 10$; (ii) $M = \left(\frac{16}{3}, \frac{15}{2}\right)$; (iii) $\frac{5\sqrt{13}}{3}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 15 (cg2-015)

Difficulty: Foundation / Marks: 7

The segment CD has midpoint $(2, -1)$, gradient $\frac{3}{4}$ and length 10. Find the coordinates of C and D .

Worked solution.

Step 1: A gradient of $\frac{3}{4}$ means the displacement from midpoint to each end is of the form $(4k, 3k)$ for some k .

$$\text{Half-displacement} = (4k, 3k)$$

The direction vector $(4, 3)$ has magnitude $\sqrt{16 + 9} = 5$.

Step 2: The full length is $2 \times 5|k| = 10$, so $|k| = 1$.

$$10|k| = 10 \implies k = 1$$

The full segment length is twice the half-length: $2 \times 5|k| = 10$, giving $|k| = 1$. We take $k = 1$ (choosing a direction).

Step 3: Find C and D using midpoint $(2, -1)$ and displacement $(4, 3)$.

$$C = (2 - 4, -1 - 3) = (-2, -4), \quad D = (2 + 4, -1 + 3) = (6, 2)$$

Subtract the displacement from the midpoint to get one endpoint, and add it to get the other.

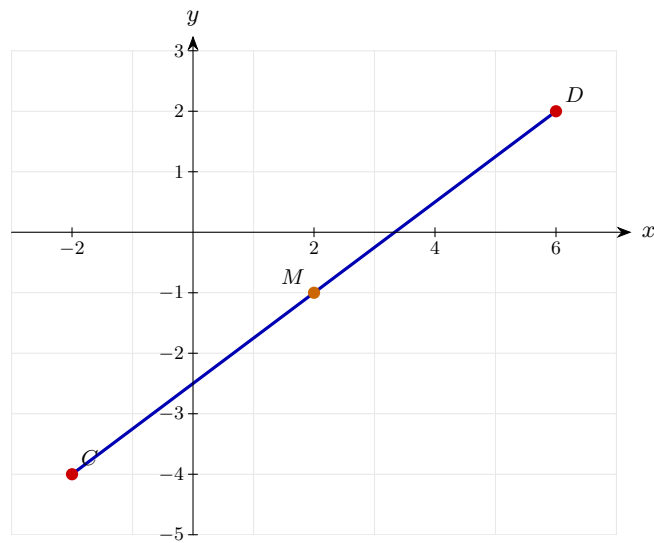
Step 4: Verify length:

$$d = \sqrt{(6 - (-2))^2 + (2 - (-4))^2} = \sqrt{64 + 36} = \sqrt{100} = 10 \checkmark$$

Always verify your answer. The length matches the given value of 10, confirming the endpoints are correct.

Final answer: $C = (-2, -4)$ and $D = (6, 2)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 16 (cg2-016)

Difficulty: Foundation / Marks: 2

Find the midpoint of $A(-3, 8)$ and $B(5, -2)$.

Worked solution.

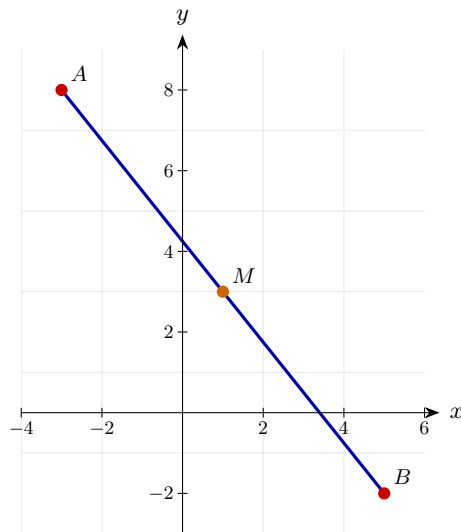
Apply midpoint formula

$$M = \left(\frac{-3 + 5}{2}, \frac{8 + (-2)}{2} \right) = (1, 3)$$

Average each coordinate.

Final answer: $(1, 3)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 17 (cg2-017)

Difficulty: Foundation / Marks: 2

Find the distance between $P(-1, 4)$ and $Q(5, -4)$.

Worked solution.

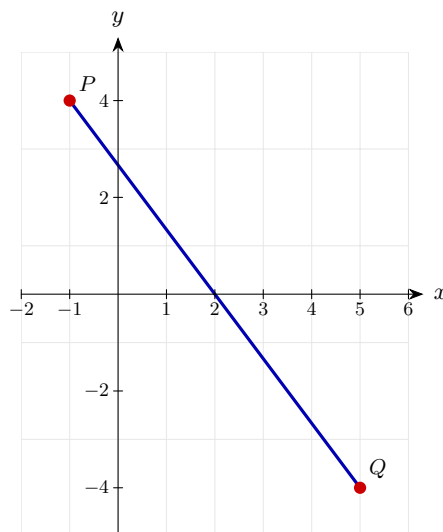
Distance formula

$$PQ = \sqrt{(5 - (-1))^2 + (-4 - 4)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$$

Using the distance formula. Be careful: $5 - (-1) = 6$, not 4. This is a 6-8-10 Pythagorean triple.

Final answer: 10

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 18 (cg2-018)

Difficulty: Foundation / Marks: 2

The midpoint of $A(2, k)$ and $B(8, 3)$ is $(5, 7)$. Find k .

Worked solution.

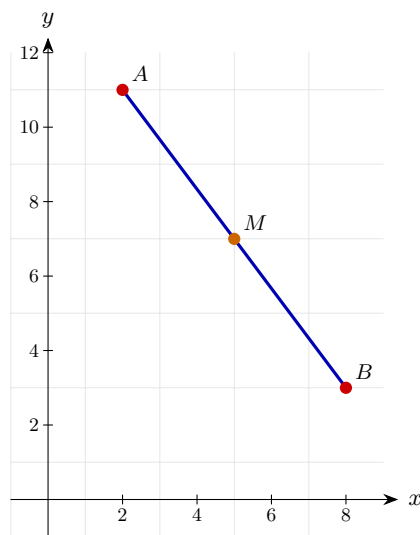
Use midpoint y-coordinate

$$\frac{k+3}{2} = 7 \implies k+3 = 14 \implies k = 11$$

Set the y-component of the midpoint formula equal to 7 and solve. Multiply both sides by 2 first to clear the fraction.

Final answer: $k = 11$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 19 (cg2-019)

Difficulty: Foundation / Marks: 2

Find the length of the line segment from $(0, 0)$ to $(7, 24)$.

Worked solution.

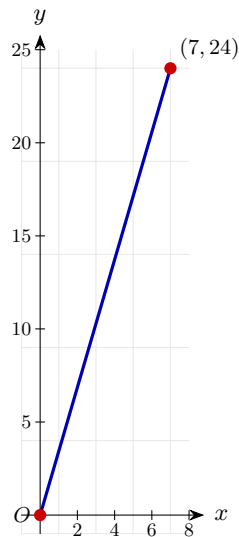
Distance formula

$$d = \sqrt{49 + 576} = \sqrt{625} = 25$$

This is a 7-24-25 Pythagorean triple. When one endpoint is the origin, the formula simplifies to $\sqrt{x^2 + y^2}$.

Final answer: 25

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 20 (cg2-020)

Difficulty: Foundation / Marks: 3

The midpoint of PQ is $(3, -1)$. If $P = (7, 5)$, find Q .

Worked solution.

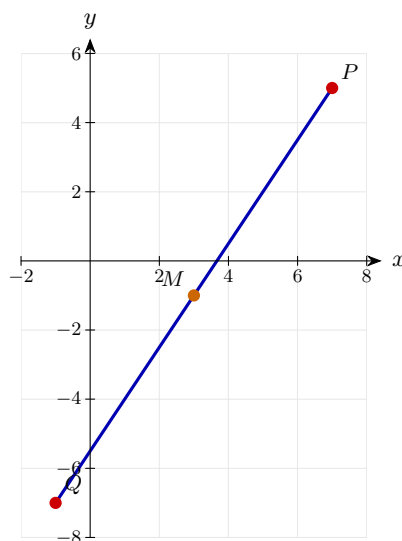
Use midpoint formula backwards

$$\frac{7 + x}{2} = 3 \implies x = -1; \quad \frac{5 + y}{2} = -1 \implies y = -7$$

Rearranging the midpoint formula for each coordinate: set $(x_P + x_Q)/2$ equal to the midpoint coordinate and solve for the unknown.

Final answer: $Q = (-1, -7)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 21 (cg2-021)

Difficulty: Foundation / Marks: 3

Show that the triangle with vertices $A(1, 1)$, $B(5, 1)$, $C(3, 5)$ is isosceles.

Worked solution.

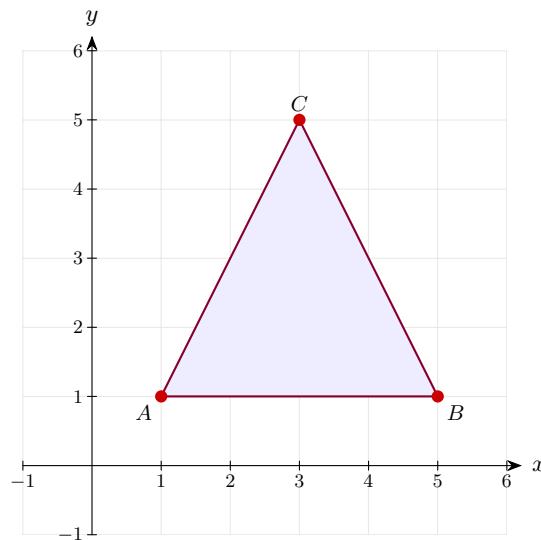
Find lengths

$$AB = 4, \quad AC = \sqrt{4 + 16} = \sqrt{20}, \quad BC = \sqrt{4 + 16} = \sqrt{20}$$

$AC = BC$ so the triangle is isosceles.

Final answer: $AC = BC = \sqrt{20}$, so triangle is isosceles.

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 22 (cg2-022)

Difficulty: Foundation / Marks: 3

Find the point that divides the line segment from $A(1, 2)$ to $B(7, 8)$ in the ratio 1 : 2.

Worked solution.

Step 1: Recall the section formula.

$$\text{If } P \text{ divides } AB \text{ in the ratio } m : n, \text{ then } P = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

The point P is closer to A since the ratio is 1:2, meaning P is one-third of the way from A to B.

Step 2: Identify the values: $A(1, 2)$, $B(7, 8)$, $m = 1$, $n = 2$.

$$x_1 = 1, \quad y_1 = 2, \quad x_2 = 7, \quad y_2 = 8$$

A is the first point and B is the second point. The ratio 1:2 means $m = 1$ (towards B) and $n = 2$ (towards A).

Step 3: Calculate the x-coordinate of P.

$$x_P = \frac{1(7) + 2(1)}{1 + 2} = \frac{7 + 2}{3} = \frac{9}{3} = 3$$

Multiply x by m and x by n, then divide by (m + n).

Step 4: Calculate the y-coordinate of P.

$$y_P = \frac{1(8) + 2(2)}{1 + 2} = \frac{8 + 4}{3} = \frac{12}{3} = 4$$

Multiply y by m and y by n, then divide by (m + n).

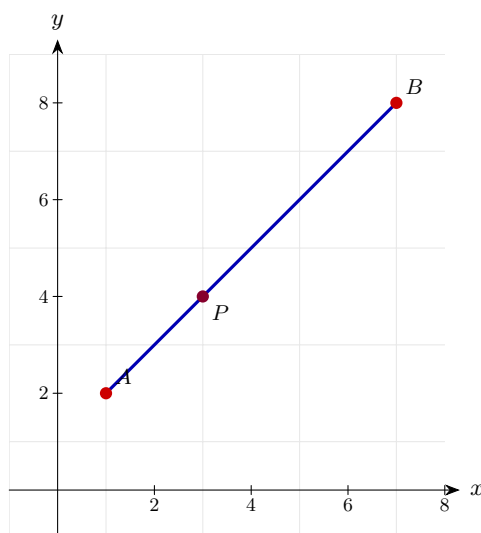
Step 5: State the answer.

$$P = (3, 4)$$

We can verify: from A(1,2) to P(3,4) the change is (+2, +2), and from P(3,4) to B(7,8) the change is (+4, +4). The ratio of these changes is 2:4 = 1:2, confirming our answer.

Final answer: $P = (3, 4)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 23 (cg2-023)

Difficulty: Foundation / Marks: 3

Find the distance between $(-3, -5)$ and $(9, 11)$. Give your answer as a surd in simplest form.

Worked solution.

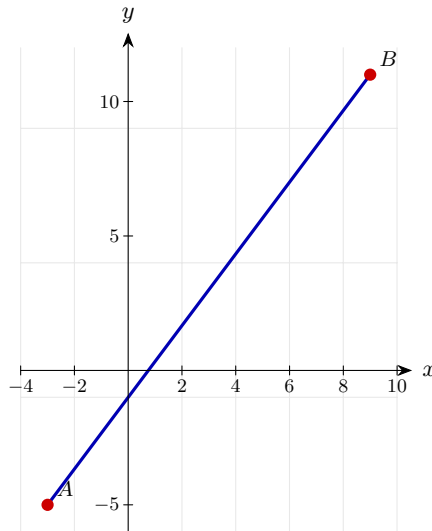
Distance formula

$$d = \sqrt{12^2 + 16^2} = \sqrt{144 + 256} = \sqrt{400} = 20$$

The differences are $9 - (-3) = 12$ and $11 - (-5) = 16$. This is a 12-16-20 triple (a multiple of 3-4-5).

Final answer: 20

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 24 (cg2-024)

Difficulty: Foundation / Marks: 4

The points $A(a, 3)$ and $B(5, a)$ are such that $AB = 5$. Find the possible values of a .

Worked solution.

Step 1: Set up distance equation

$$(5 - a)^2 + (a - 3)^2 = 25$$

Apply the distance formula and square both sides to eliminate the square root, since $AB = 5$ means $AB^2 = 25$.

Step 2: Expand and simplify

$$25 - 10a + a^2 + a^2 - 6a + 9 = 25 \implies 2a^2 - 16a + 9 = 0$$

Expand each squared bracket carefully and collect like terms. The 25 on each side cancels.

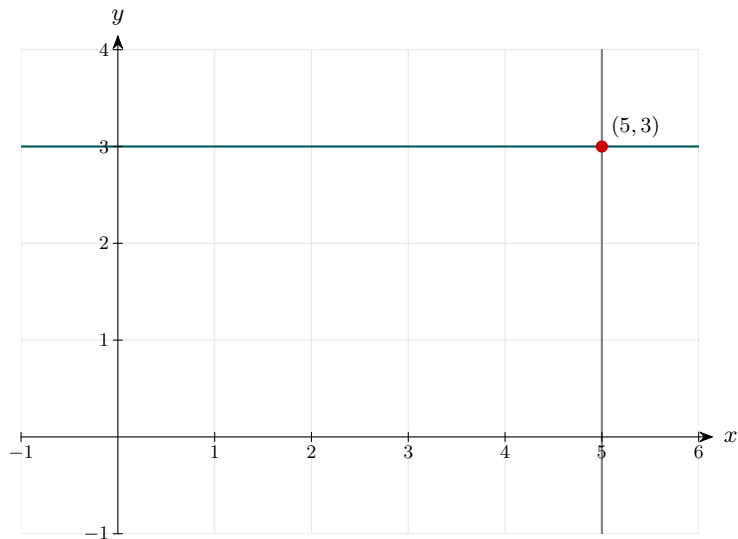
Step 3: Solve using quadratic formula

$$a = \frac{16 \pm \sqrt{256 - 72}}{4} = \frac{16 \pm \sqrt{184}}{4} = 4 \pm \frac{\sqrt{46}}{2}$$

Apply the quadratic formula with $a=2$, $b=-16$, $c=9$. Two solutions exist because there are two points at distance 5 from $B(5, a)$ with y -coordinate 3.

Final answer: $a = 4 \pm \frac{\sqrt{46}}{2}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 25 (cg2-025)

Difficulty: Foundation / Marks: 2

Find the midpoint of the line segment joining $(2a, 3b)$ and $(6a, -b)$.

Worked solution.

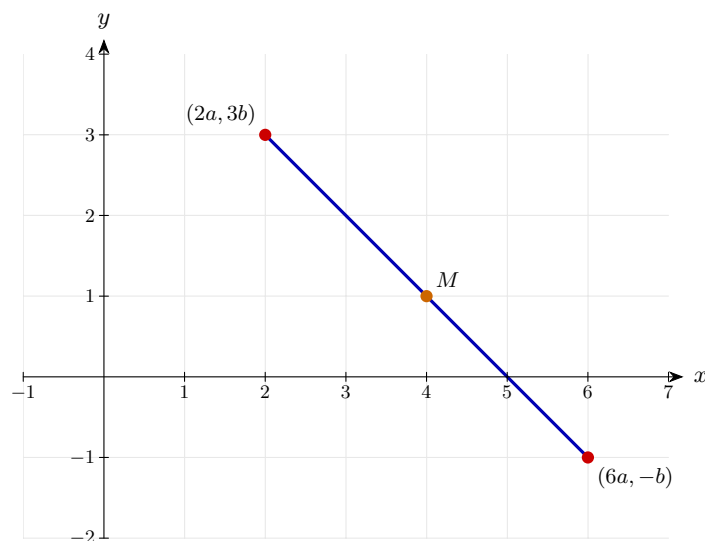
Midpoint formula

$$M = \left(\frac{2a + 6a}{2}, \frac{3b + (-b)}{2} \right) = (4a, b)$$

The midpoint formula works with algebraic expressions just as with numbers. Simplify: $8a/2 = 4a$ and $2b/2 = b$.

Final answer: $(4a, b)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 26 (cg2-026)

Difficulty: Foundation / Marks: 4

Show that $A(0,0)$, $B(3,4)$, $C(8,4)$, $D(5,0)$ form a parallelogram by showing the diagonals bisect each other.

Worked solution.

Step 1: Identify the diagonals of quadrilateral ABCD.

Diagonals: AC and BD

In a quadrilateral ABCD, the diagonals connect opposite vertices: A to C and B to D. A key property of a parallelogram is that its diagonals bisect each other — they share the same midpoint.

Step 2: Find the midpoint of diagonal AC, where $A(0,0)$ and $C(8,4)$.

$$M_{AC} = \left(\frac{0+8}{2}, \frac{0+4}{2} \right) = (4, 2)$$

Using the midpoint formula: add the x-coordinates and divide by 2, then add the y-coordinates and divide by 2.

Step 3: Find the midpoint of diagonal BD, where $B(3,4)$ and $D(5,0)$.

$$M_{BD} = \left(\frac{3+5}{2}, \frac{4+0}{2} \right) = (4, 2)$$

Again applying the midpoint formula to the other diagonal.

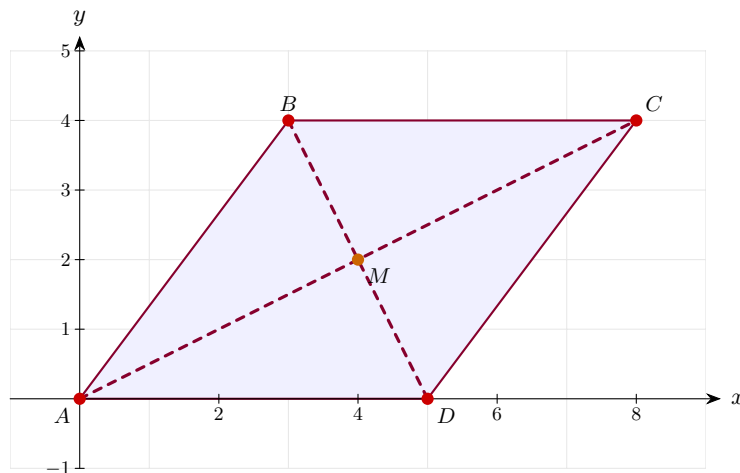
Step 4: Compare the two midpoints.

$$M_{AC} = (4, 2) = M_{BD}$$

Both diagonals have the same midpoint $(4, 2)$, which means they bisect each other. This is the defining property of a parallelogram, so ABCD is a parallelogram.

Final answer: Diagonals AC and BD both have midpoint $(4,2)$, so they bisect each other, proving ABCD is a parallelogram.

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 27 (cg2-027)

Difficulty: Foundation / Marks: 3

Find the perimeter of the triangle with vertices $A(0,0)$, $B(4,0)$, $C(0,3)$.

Worked solution.

Step 1: Find all side lengths

$$AB = 4, \quad AC = 3, \quad BC = \sqrt{16 + 9} = 5$$

AB and AC lie along the axes, so their lengths are just the coordinate differences. Only BC requires the full distance formula. This is a 3-4-5 right triangle.

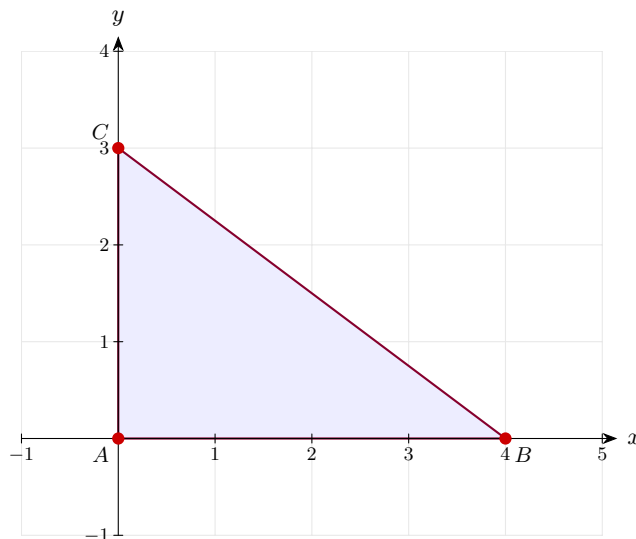
Step 2: Sum

$$P = 4 + 3 + 5 = 12$$

The perimeter is the sum of all three side lengths.

Final answer: 12

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 28 (cg2-028)

Difficulty: Foundation / Marks: 3

The point P divides AB in the ratio $3 : 1$, where $A = (2, -1)$ and $B = (10, 7)$. Find P .

Worked solution.

Step 1: Recall the section formula.

$$\text{If } P \text{ divides } AB \text{ in the ratio } m : n, \text{ then } P = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

The ratio 3:1 means P is three-quarters of the way from A to B , so P is much closer to B than to A .

Step 2: Identify the values: $A(2, -1)$, $B(10, 7)$, $m = 3$, $n = 1$.

$$x_1 = 2, \quad y_1 = -1, \quad x_2 = 10, \quad y_2 = 7$$

Here $m = 3$ (towards B) and $n = 1$ (towards A), and the total parts are $m + n = 4$.

Step 3: Calculate the x -coordinate of P .

$$x_P = \frac{3(10) + 1(2)}{3 + 1} = \frac{30 + 2}{4} = \frac{32}{4} = 8$$

Multiply x by m and x by n , then divide by the total number of parts.

Step 4: Calculate the y -coordinate of P .

$$y_P = \frac{3(7) + 1(-1)}{3 + 1} = \frac{21 - 1}{4} = \frac{20}{4} = 5$$

Same process for the y -coordinates.

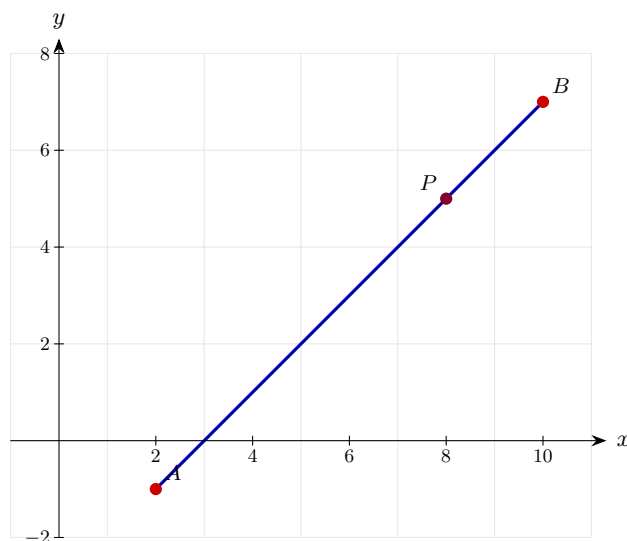
Step 5: State the answer.

$$P = (8, 5)$$

We can verify: from $A(2, -1)$ to $P(8, 5)$ the change is $(+6, +6)$, and from $P(8, 5)$ to $B(10, 7)$ the change is $(+2, +2)$. The ratio $6:2 = 3:1$, confirming our answer.

Final answer: $P = (8, 5)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 29 (cg2-029)

Difficulty: Foundation / Marks: 2

Find the distance from the origin to the point $(5, -12)$.

Worked solution.

Step 1: Recall the distance formula.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The origin is the point (0, 0), so we use A(0, 0) and B(5, -12).

Step 2: Substitute the coordinates.

$$d = \sqrt{(5 - 0)^2 + (-12 - 0)^2} = \sqrt{5^2 + (-12)^2}$$

Subtract the x-coordinates and the y-coordinates separately, then square each difference.

Step 3: Evaluate the squares and add.

$$d = \sqrt{25 + 144} = \sqrt{169}$$

Note that $(-12)^2 = 144$, not -144. Squaring always gives a positive result.

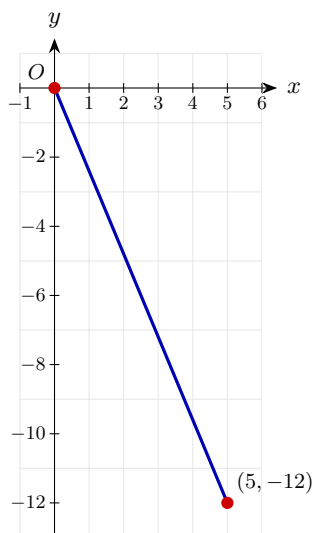
Step 4: Simplify the square root.

$$d = 13$$

$169 = 13 \times 13$, so $\sqrt{169} = 13$. This is a 5-12-13 Pythagorean triple.

Final answer: 13

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 30 (cg2-030)

Difficulty: Foundation / Marks: 4

The endpoints of a diameter of a circle are A(-2, 3) and B(6, -1). Find the centre and radius of the circle.

Worked solution.

Step 1: Centre = midpoint

$$C = \left(\frac{-2 + 6}{2}, \frac{3 + (-1)}{2} \right) = (2, 1)$$

The centre of a circle lies at the midpoint of any diameter.

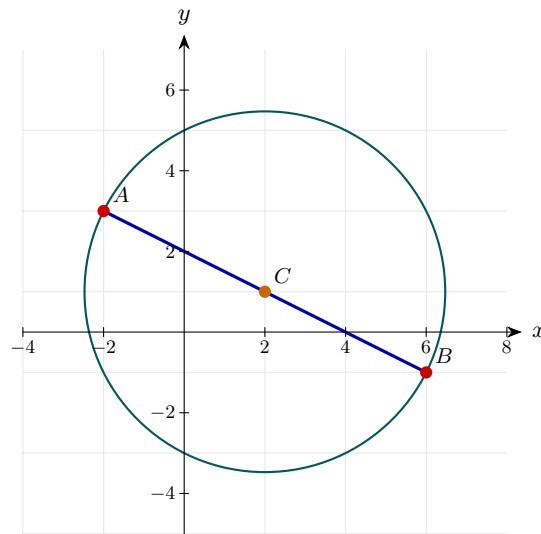
Step 2: Radius = half diameter length

$$AB = \sqrt{64 + 16} = \sqrt{80} = 4\sqrt{5}, \quad r = 2\sqrt{5}$$

Find the full diameter length using the distance formula, then halve it. Simplify: $\sqrt{80} = \sqrt{16 \times 5} = 4\sqrt{5}$.

Final answer: Centre $(2, 1)$, radius $2\sqrt{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 31 (cg2-031)

Difficulty: Foundation / Marks: 4

Prove that the triangle with vertices $(1, 1)$, $(4, 5)$, $(7, 1)$ is isosceles but not equilateral.

Worked solution.

Step 1: Find all lengths

$$AB = \sqrt{9 + 16} = 5, \quad BC = \sqrt{9 + 16} = 5, \quad AC = 6$$

Use the distance formula for each pair. AC is horizontal (same y-coordinates), so its length is simply the difference in x-coordinates.

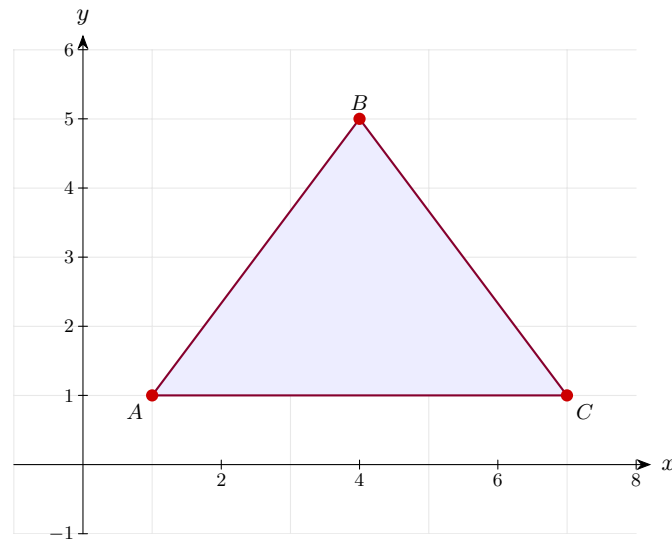
Step 2: Conclusion

$$AB = BC = 5 \neq AC = 6$$

Two sides equal so isosceles, but not all equal so not equilateral.

Final answer: $AB = BC = 5$, $AC = 6$. Isosceles but not equilateral.

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 32 (cg2-032)

Difficulty: Foundation / Marks: 4

The midpoint of AB is $M(4, -2)$ and $A = (1, 3)$. Find B and the length AB .

Worked solution.

Step 1: Find B

$$B = (2(4) - 1, 2(-2) - 3) = (7, -7)$$

Rearranging the midpoint formula: $x_B = 2x_M - x_A$ and $y_B = 2y_M - y_A$.

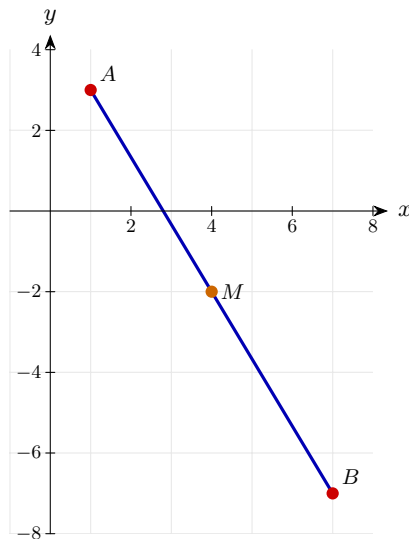
Step 2: Length

$$AB = \sqrt{36 + 100} = \sqrt{136} = 2\sqrt{34}$$

Apply the distance formula between $A(1, 3)$ and $B(7, -7)$. Simplify: $\sqrt{136} = \sqrt{4 \cdot 34} = 2\sqrt{34}$.

Final answer: $B = (7, -7), AB = 2\sqrt{34}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 33 (cg2-033)

Difficulty: Foundation / Marks: 2

Find the point that divides $A(0,0)$ to $B(12,8)$ in the ratio 2 : 1.

Worked solution.

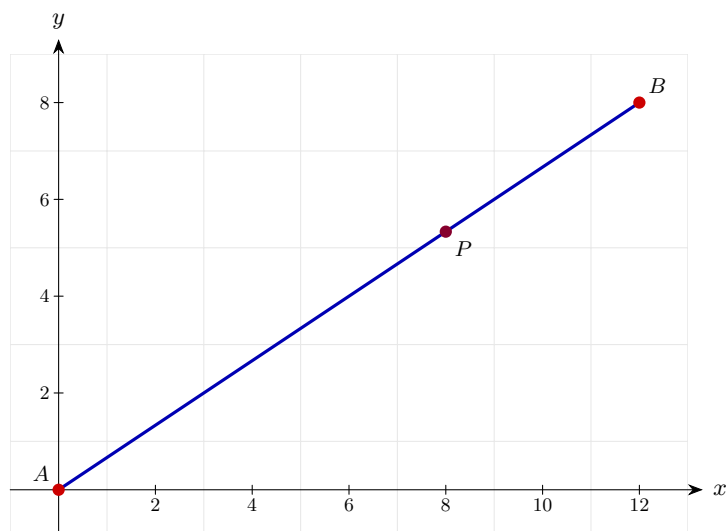
Section formula

$$P = \left(\frac{2(12)}{3}, \frac{2(8)}{3} \right) = \left(8, \frac{16}{3} \right)$$

The ratio 2:1 means P is two-thirds of the way from A to B. Since A is the origin, each coordinate of P is simply $\frac{2}{3}$ of the corresponding coordinate of B.

Final answer: $\left(8, \frac{16}{3} \right)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 34 (cg2-034)

Difficulty: Foundation / Marks: 4

Show that the points $A(2, 3)$, $B(5, 7)$, $C(9, 4)$ form a right-angled triangle at B .

Worked solution.

Step 1: Find lengths squared

$$AB^2 = 3^2 + 4^2 = 25, \quad BC^2 = 4^2 + (-3)^2 = 25, \quad AC^2 = 7^2 + 1^2 = 50$$

Working with squared lengths avoids square roots and lets us check Pythagoras directly.

Step 2: Check Pythagoras

$$AB^2 + BC^2 = 25 + 25 = 50 = AC^2$$

Since the two shorter sides meet at B and their squares add to the square of AC , the right angle is at B .

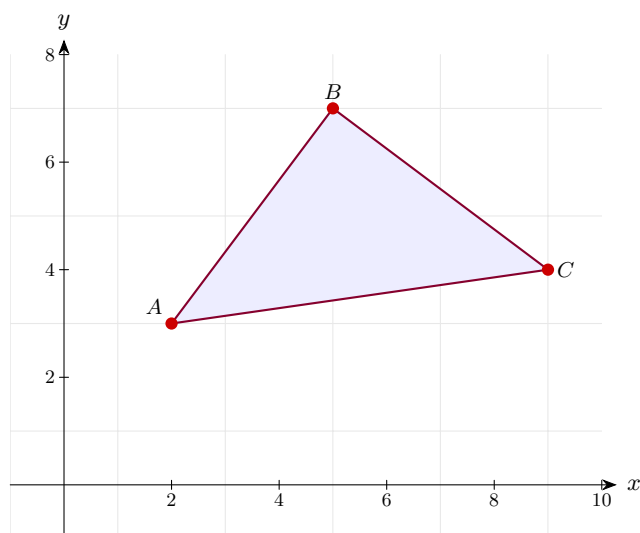
Step 3: Check perpendicular gradients

$$m_{AB} = \frac{4}{3}, \quad m_{BC} = \frac{-3}{4}, \quad m_{AB} \times m_{BC} = -1$$

The gradients are negative reciprocals, so AB is perpendicular to BC . This confirms the right angle at B .

Final answer: Right-angled at B , with $AB = BC = 5$ and $AC = 5\sqrt{2}$.

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 35 (cg2-035)

Difficulty: Foundation / Marks: 2

Find the distance between $A(1, -2)$ and $B(-3, 1)$. Leave your answer as a surd.

Worked solution.

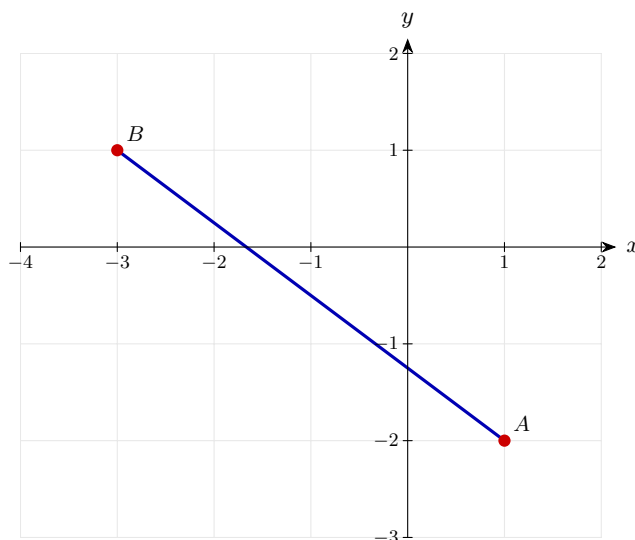
Distance formula

$$AB = \sqrt{(-3-1)^2 + (1-(-2))^2} = \sqrt{16+9} = 5$$

Note that $(-3-1)^2 = 16$ and $(1-(-2))^2 = 9$. The answer is a whole number (3-4-5 triple), so no surd is needed despite the question asking for one.

Final answer: 5

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 36 (cg2-036)

Difficulty: Foundation / Marks: 5

M is the midpoint of AB where $A = (-4, 6)$ and $B = (2, -8)$. Find the gradient of AB and the equation of the perpendicular bisector.

Worked solution.

Step 1: Midpoint

$$M = (-1, -1)$$

Using the midpoint formula: average the x-coordinates and average the y-coordinates separately.

Step 2: Gradient of AB

$$m = \frac{-8-6}{2-(-4)} = \frac{-14}{6} = -\frac{7}{3}$$

Gradient = rise over run. Be careful with double negatives: $2 - (-4) = 6$, not 2.

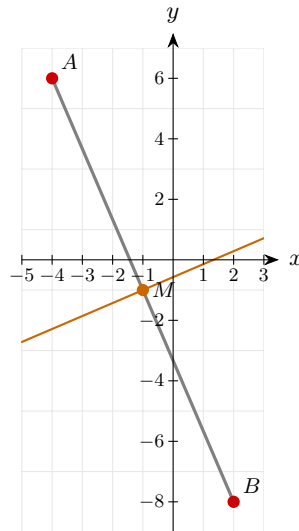
Step 3: Perpendicular bisector

$$m_{\perp} = \frac{3}{7}; \quad y + 1 = \frac{3}{7}(x + 1) \implies 7y = 3x - 4$$

The perpendicular gradient is the negative reciprocal. Then use $y - y_1 = m(x - x_1)$ with the midpoint as (x_1, y_1) .

Final answer: Gradient of AB : $-\frac{7}{3}$; perpendicular bisector: $3x - 7y - 4 = 0$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 37 (cg2-037)

Difficulty: Foundation | Marks: 5

The three vertices of a triangle are $A(0,0)$, $B(8,0)$ and $C(4,6)$. Find the lengths of the three medians.

Worked solution.

Step 1: Midpoints of sides

$$M_{BC} = (6, 3), \quad M_{AC} = (2, 3), \quad M_{AB} = (4, 0)$$

A median connects a vertex to the midpoint of the opposite side, so first find all three midpoints.

Step 2: Median from A

$$AM_{BC} = \sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5}$$

Distance from $A(0,0)$ to $M_{BC}(6,3)$. Simplify: $\sqrt{45} = \sqrt{9 \times 5} = 3\sqrt{5}$.

Step 3: Median from B

$$BM_{AC} = \sqrt{36 + 9} = 3\sqrt{5}$$

Distance from $B(8,0)$ to $M_{AC}(2,3)$. Two medians have equal length, reflecting the symmetry of the triangle.

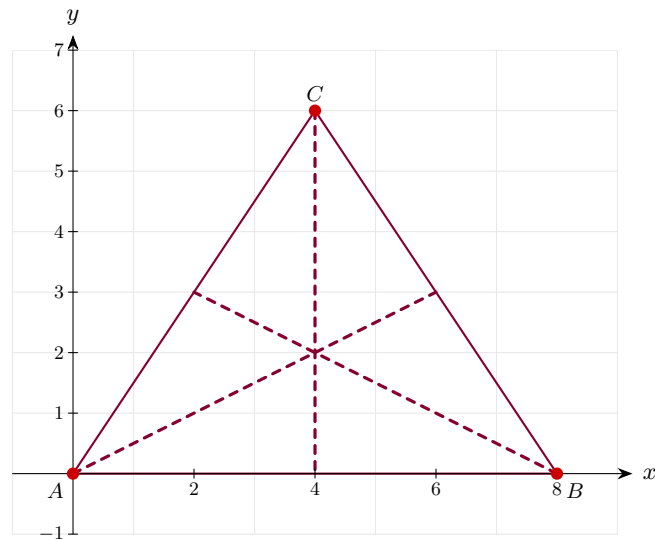
Step 4: Median from C

$$CM_{AB} = \sqrt{0 + 36} = 6$$

Distance from $C(4,6)$ to $M_{AB}(4,0)$. Since they share the same x-coordinate, the distance is simply the difference in y-coordinates.

Final answer: $3\sqrt{5}, 3\sqrt{5}, 6$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 38 (cg2-038)

Difficulty: Foundation / Marks: 3

A line segment has endpoints $(1, 5)$ and $(9, 1)$. Find the point that is $\frac{3}{4}$ of the way from the first to the second endpoint.

Worked solution.

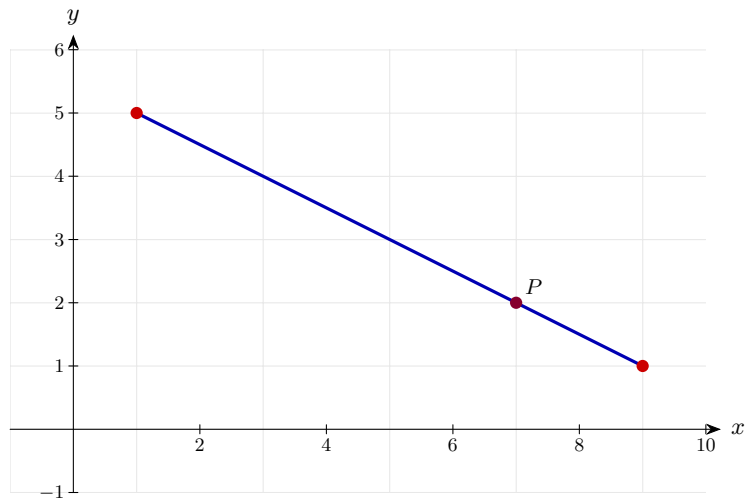
Use ratio 3:1

$$P = \left(1 + \frac{3}{4}(9 - 1), 5 + \frac{3}{4}(1 - 5) \right) = (7, 2)$$

Being $\frac{3}{4}$ of the way along is equivalent to dividing the segment in the ratio 3:1. Add $\frac{3}{4}$ of the change in each coordinate to the starting point.

Final answer: $(7, 2)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 39 (cg2-039)

Difficulty: Foundation / *Marks: 4*

The distance from $(k, 2k)$ to $(3, 1)$ is $\sqrt{13}$. Find the possible values of k .

Worked solution.

Step 1: Set up equation

$$(k - 3)^2 + (2k - 1)^2 = 13$$

Apply the distance formula and square both sides to remove the square root. Since $d = \sqrt{13}$, squaring gives $d^2 = 13$.

Step 2: Expand

$$k^2 - 6k + 9 + 4k^2 - 4k + 1 = 13 \implies 5k^2 - 10k - 3 = 0$$

Expand each bracket carefully, then collect like terms and move everything to one side.

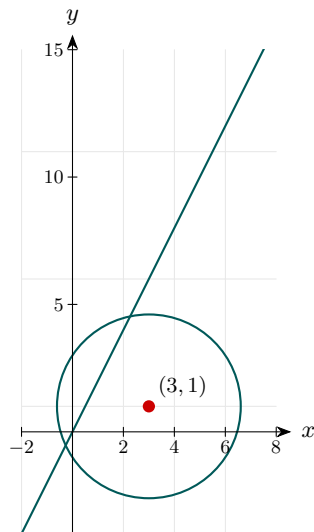
Step 3: Solve

$$k = \frac{10 \pm \sqrt{100 + 60}}{10} = 1 \pm \frac{2\sqrt{10}}{5}$$

Use the quadratic formula. Two solutions arise because two points at distance $\sqrt{13}$ from $(3, 1)$ lie on the line $y = 2x$.

Final answer: $k = 1 \pm \frac{2\sqrt{10}}{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 40 (cg2-040)

Difficulty: Foundation / Marks: 2

Find the coordinates of the centroid of the triangle with vertices (2, 1), (8, 3), (5, 8).

Worked solution.

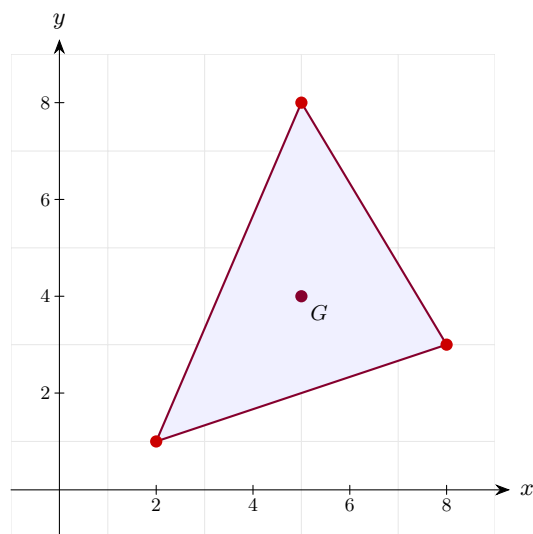
Centroid formula

$$G = \left(\frac{2 + 8 + 5}{3}, \frac{1 + 3 + 8}{3} \right) = (5, 4)$$

The centroid is the mean of the three vertices: average all x-coordinates and average all y-coordinates. It is the point where all three medians intersect.

Final answer: (5, 4)

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 41 (cg2-041)

Difficulty: Foundation / Marks: 4

The points $A(1, 2)$ and $B(7, 10)$ are the endpoints of a diameter of a circle. Find the equation of the circle.

Worked solution.

Step 1: Centre

$$C = (4, 6)$$

Midpoint of AB.

Step 2: Radius

$$r = \frac{1}{2}\sqrt{36 + 64} = \frac{1}{2}(10) = 5$$

The radius is half the diameter length. Find the full distance AB first, then halve it.

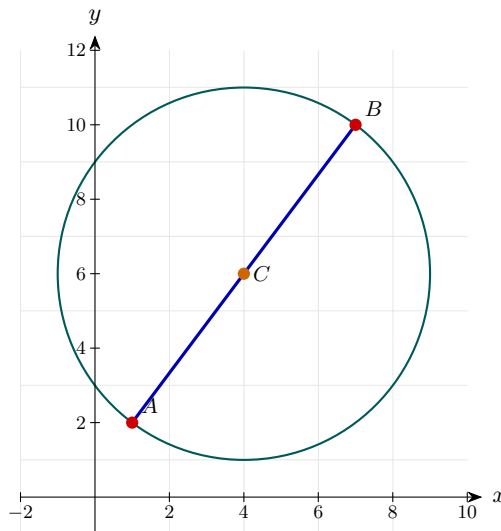
Step 3: Equation

$$(x - 4)^2 + (y - 6)^2 = 25$$

The standard circle equation is $(x - a)^2 + (y - b)^2 = r^2$, where (a, b) is the centre. Remember to write r^2 , not r , on the right side.

Final answer: $(x - 4)^2 + (y - 6)^2 = 25$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 42 (cg2-042)

Difficulty: Foundation / Marks: 3

Find the distance between the parallel lines $y = 2x + 1$ and $y = 2x + 6$.

Worked solution.

Step 1: Rewrite as $ax + by + c = 0$

$$2x - y + 1 = 0 \text{ and } 2x - y + 6 = 0$$

Both lines must be in the form $ax + by + c = 0$ with the same a and b values to use the parallel lines distance formula.

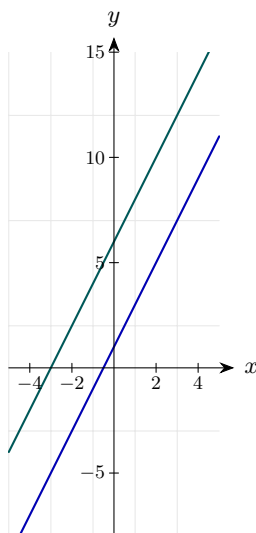
Step 2: Distance between parallel lines

$$d = \frac{|6 - 1|}{\sqrt{4 + 1}} = \frac{5}{\sqrt{5}} = \sqrt{5}$$

The formula for distance between parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is $\frac{|c_2 - c_1|}{\sqrt{a^2 + b^2}}$.
Rationalise: $\frac{5}{\sqrt{5}} = \sqrt{5}$.

Final answer: $\sqrt{5}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 43 (cg2-043)

Difficulty: Foundation / Marks: 3

A quadrilateral has vertices $A(0,0)$, $B(5,0)$, $C(7,4)$, $D(2,4)$. Find the lengths of its diagonals.

Worked solution.

Step 1: Diagonal AC

$$AC = \sqrt{49 + 16} = \sqrt{65}$$

Distance from $A(0,0)$ to $C(7,4)$. Since 65 has no perfect square factors other than 1, $\sqrt{65}$ is already in simplest surd form.

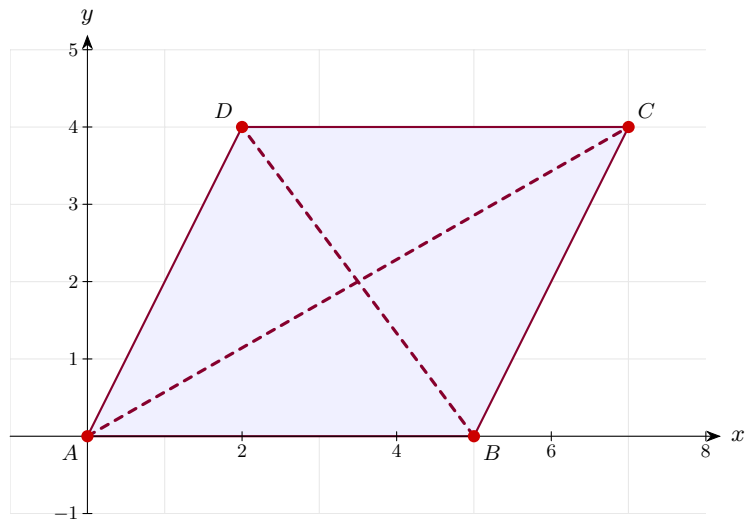
Step 2: Diagonal BD

$$BD = \sqrt{9 + 16} = 5$$

Distance from $B(5,0)$ to $D(2,4)$. This is a 3-4-5 Pythagorean triple.

Final answer: $AC = \sqrt{65}$, $BD = 5$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 44 (cg2-044)

Difficulty: Foundation / Marks: 3

The point P divides AB externally in the ratio $3 : 1$, where $A = (2, 4)$ and $B = (6, 8)$. Find P .

Worked solution.

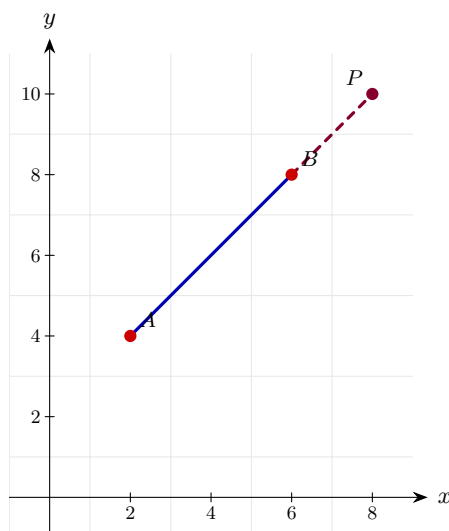
External division formula

$$P = \left(\frac{3(6) - 1(2)}{3 - 1}, \frac{3(8) - 1(4)}{3 - 1} \right) = (8, 10)$$

For external division in ratio $m : n$, use the formula $P = \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n} \right)$. Note the minus signs — this is the key difference from internal division.

Final answer: $(8, 10)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 45 (cg2-045)

Difficulty: Foundation / Marks: 2

The midpoint of AB is $(0, 0)$. If $A = (3a, -2b)$, express B in terms of a and b .

Worked solution.

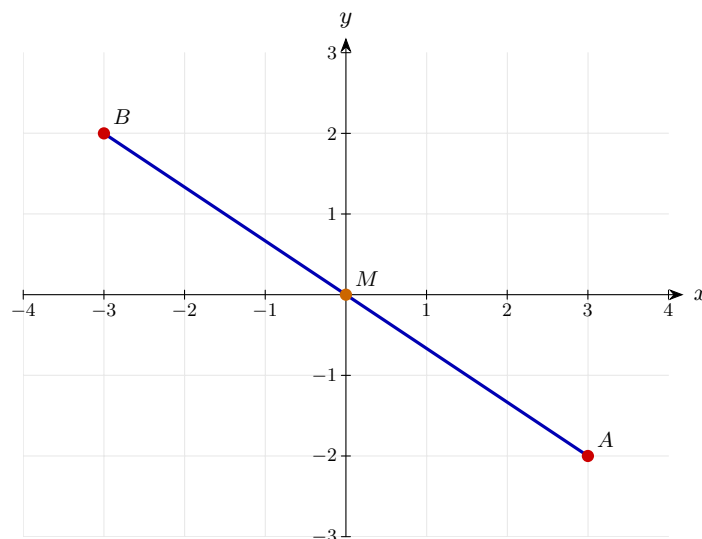
Use midpoint $= (0, 0)$

$$B = (-3a, 2b)$$

Since midpoint is origin, B is the negative of A .

Final answer: $B = (-3a, 2b)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 46 (cg2-046)

Difficulty: Foundation / Marks: 2

Find the area of the triangle with vertices $(0, 0)$, $(6, 0)$, $(3, 8)$.

Worked solution.

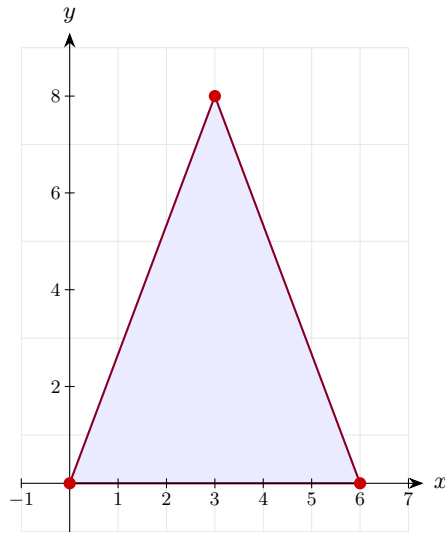
Area formula

$$\text{Area} = \frac{1}{2} \times 6 \times 8 = 24$$

Base $= 6$ along x-axis, height $= 8$.

Final answer: 24 square units

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 47 (cg2-047)

Difficulty: Foundation / Marks: 5

Show that $A(-1, 2)$, $B(3, 0)$, $C(5, 4)$, $D(1, 6)$ form a square.

Worked solution.

Step 1: Find all side lengths

$$AB = \sqrt{16 + 4} = \sqrt{20}, \quad BC = \sqrt{4 + 16} = \sqrt{20}$$

Apply the distance formula to each pair of consecutive vertices. To show a square, we need all four sides equal and at least one right angle.

Step 2: Remaining sides

$$CD = \sqrt{16 + 4} = \sqrt{20}, \quad DA = \sqrt{4 + 16} = \sqrt{20}$$

All sides equal.

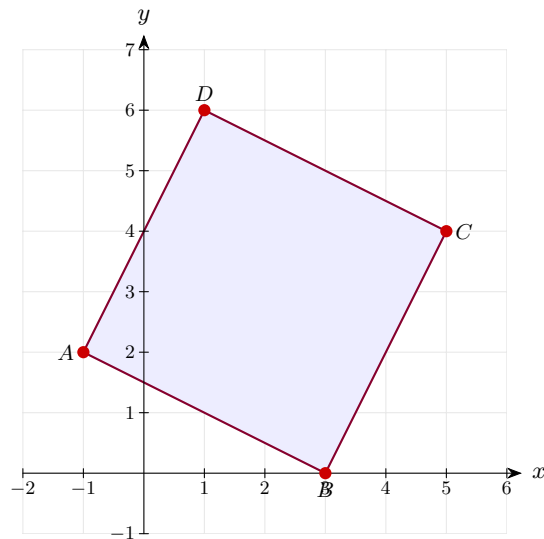
Step 3: Check right angle

$$m_{AB} = -\frac{1}{2}, \quad m_{BC} = 2; \quad m_{AB} \times m_{BC} = -1 \checkmark$$

Perpendicular adjacent sides.

Final answer: All sides $\sqrt{20}$ and adjacent sides perpendicular, so ABCD is a square.

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 48 (cg2-048)

Difficulty: Foundation / Marks: 3

Find the length of the line segment from $(\cos \theta, \sin \theta)$ to $(-\cos \theta, -\sin \theta)$.

Worked solution.

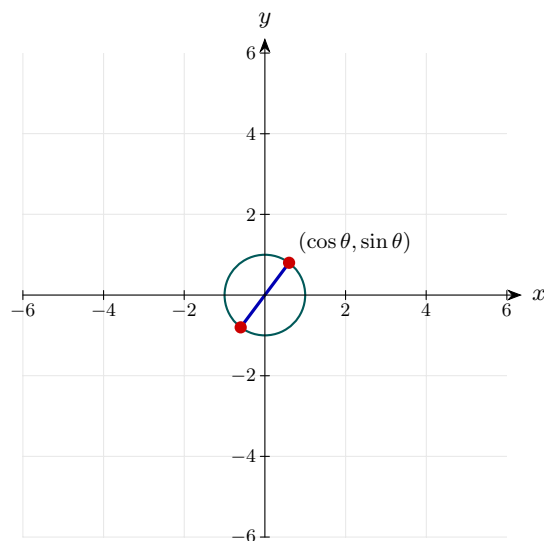
Distance formula

$$d = \sqrt{(2 \cos \theta)^2 + (2 \sin \theta)^2} = 2\sqrt{\cos^2 \theta + \sin^2 \theta} = 2$$

Using the Pythagorean identity.

Final answer: 2

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 49 (cg2-049)

Difficulty: Foundation / Marks: 4

The line segment AB has length 10. $A = (1, 2)$ and B lies on the x -axis. Find the possible coordinates of B .

Worked solution.

Step 1: Let $B = (x, 0)$

$$(x - 1)^2 + 4 = 100 \implies (x - 1)^2 = 96$$

Since B is on the x -axis, its y -coordinate is 0. Substitute into the distance formula squared: $(x - 1)^2 + (0 - 2)^2 = 10^2$.

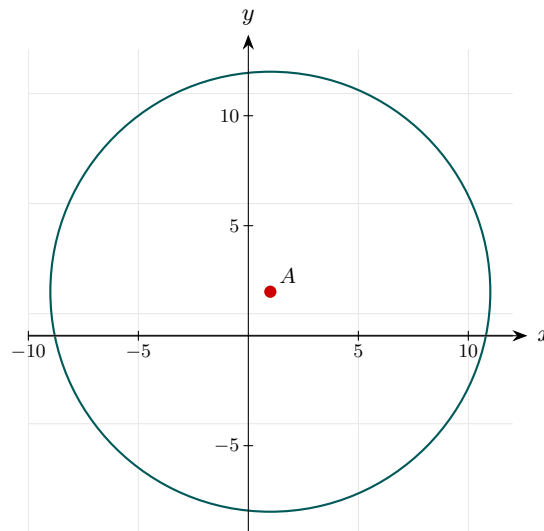
Step 2: Solve

$$x = 1 \pm 4\sqrt{6}$$

Take the square root of both sides: $\sqrt{96} = \sqrt{16 \times 6} = 4\sqrt{6}$. Two solutions arise since B could be on either side of A .

Final answer: $B = (1 + 4\sqrt{6}, 0)$ or $B = (1 - 4\sqrt{6}, 0)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 50 (cg2-050)

Difficulty: Foundation / Marks: 7

A triangle has vertices $P(2, 1)$, $Q(8, 1)$, $R(5, 7)$. Find: (a) the perimeter; (b) the area; (c) the coordinates of the centroid.

Worked solution.

Step 1: (a) Side lengths

$$PQ = 6, \quad PR = \sqrt{9 + 36} = \sqrt{45} = 3\sqrt{5}, \quad QR = \sqrt{9 + 36} = 3\sqrt{5}$$

PQ is horizontal (same y-coordinate), so its length is simply $|8 - 2| = 6$. Simplify surds: $\sqrt{45} = \sqrt{9 \times 5} = 3\sqrt{5}$.

Step 2: Perimeter

$$P = 6 + 6\sqrt{5} \approx 19.42$$

Sum all three sides: $6 + 3\sqrt{5} + 3\sqrt{5} = 6 + 6\sqrt{5}$. Collect like surds before adding.

Step 3: (b) Area

$$\text{Area} = \frac{1}{2} \times 6 \times 6 = 18$$

Base PQ = 6, height = 7 - 1 = 6.

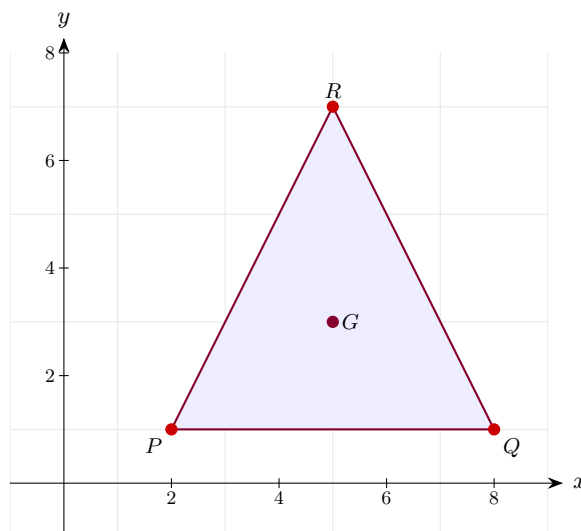
Step 4: (c) Centroid

$$G = \left(\frac{2 + 8 + 5}{3}, \frac{1 + 1 + 7}{3} \right) = (5, 3)$$

The centroid is the average of the three vertices: add all x-coordinates and divide by 3, then do the same for y-coordinates.

Final answer: (a) $6 + 6\sqrt{5}$; (b) 18; (c) (5, 3)

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 51 (cg2-051)

Difficulty: Challenge / Marks: 4

The points $A(2, 5)$, $B(8, 1)$ and $C(k, 9)$ are such that $AB = AC$. Find the possible values of k .

Worked solution.

Step 1: Set $AB = AC$ and square both sides to avoid square roots.

$$AB^2 = (8 - 2)^2 + (1 - 5)^2 = 36 + 16 = 52$$

$$AC^2 = (k - 2)^2 + (9 - 5)^2 = (k - 2)^2 + 16$$

Squaring both sides is valid since distances are positive.

Step 2: Set $AB^2 = AC^2$ and solve for k .

$$(k - 2)^2 + 16 = 52$$

$$(k - 2)^2 = 36$$

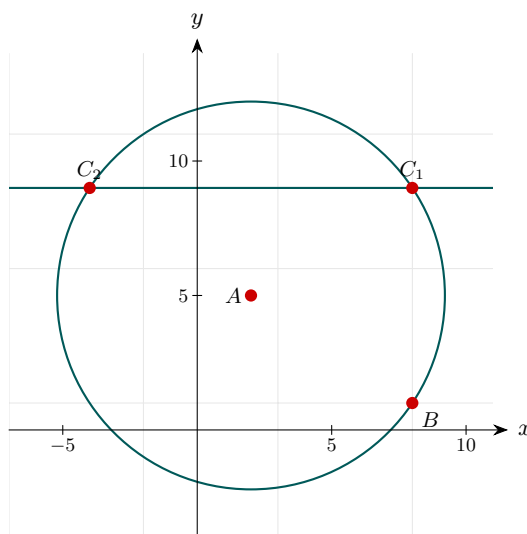
$$k - 2 = \pm 6$$

$$k = 8 \text{ or } k = -4$$

Take the square root of both sides, remembering both positive and negative roots.

Final answer: $k = 8$ or $k = -4$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 52 (cg2-052)

Difficulty: Challenge / Marks: 5

The line segment AB has midpoint $M(3, 7)$. The point A has coordinates $(-1, 2)$. The point C divides AB in the ratio $1 : 3$. Find the coordinates of C .

Worked solution.

Step 1: Use the midpoint to find B .

$$\frac{-1 + x_B}{2} = 3 \implies x_B = 7$$

$$\frac{2 + y_B}{2} = 7 \implies y_B = 12$$

Rearrange the midpoint formula to find B .

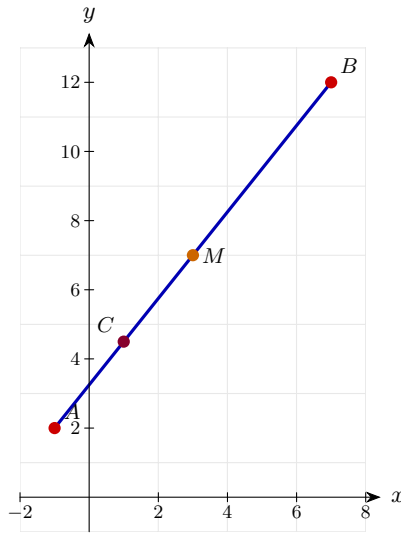
Step 2: C divides AB in the ratio $1 : 3$, so C is $\frac{1}{4}$ of the way from A to B .

$$\begin{aligned} C &= A + \frac{1}{4}(B - A) \\ &= (-1, 2) + \frac{1}{4}(8, 10) = (1, 4.5) \end{aligned}$$

The ratio $1 : 3$ means C is $\frac{1}{4}$ of the way along.

Final answer: $C = (1, 4.5)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 53 (cg2-053)

Difficulty: Challenge / Marks: 5

Show that the triangle with vertices $A(1, 1)$, $B(4, 5)$ and $C(7, 1)$ is isosceles, and find the area.

Worked solution.

Step 1: Calculate all three side lengths.

$$AB = \sqrt{9 + 16} = 5$$

$$BC = \sqrt{9 + 16} = 5$$

$$AC = 6$$

$AB = BC = 5$, so the triangle is isosceles.

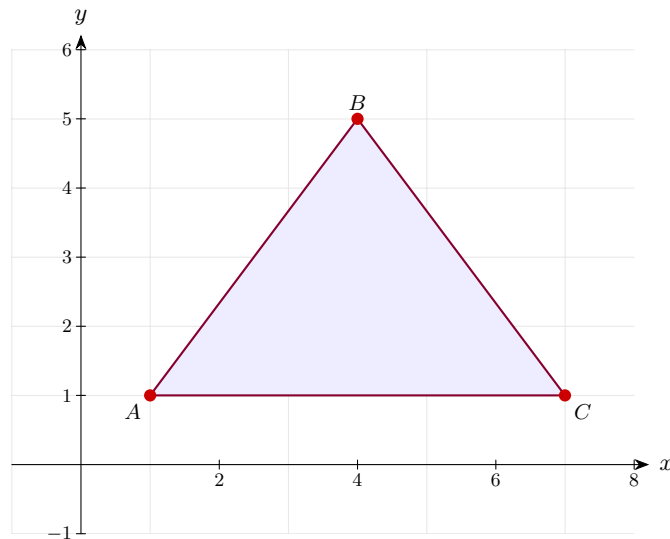
Step 2: Find the area.

$$\text{Area} = \frac{1}{2} \times 6 \times 4 = 12$$

Base $AC = 6$ is horizontal; height is $5 - 1 = 4$.

Final answer: Isosceles ($AB = BC = 5$); area = 12 square units

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 54 (cg2-054)

Difficulty: Challenge / Marks: 4

The distance from $(a, 2a)$ to $(3, -1)$ is $\sqrt{50}$. Find the possible values of a .

Worked solution.

Step 1: Use the distance formula and square both sides.

$$(a - 3)^2 + (2a + 1)^2 = 50$$

$$5a^2 - 2a + 10 = 50$$

$$5a^2 - 2a - 40 = 0$$

Expand and collect into a quadratic.

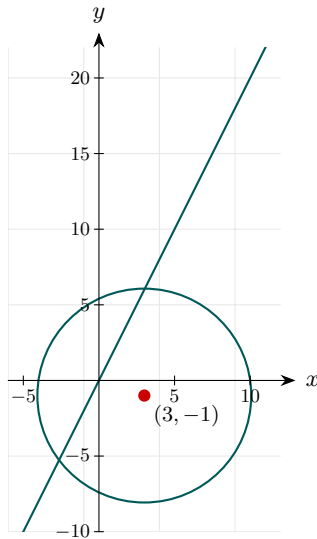
Step 2: Solve the quadratic.

$$a = \frac{2 \pm \sqrt{4 + 800}}{10} = \frac{1 \pm \sqrt{201}}{5}$$

Apply the quadratic formula.

$$\text{Final answer: } a = \frac{1 + \sqrt{201}}{5} \text{ or } a = \frac{1 - \sqrt{201}}{5}$$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 55 (cg2-055)

Difficulty: Challenge / Marks: 4

The points $P(0, 3)$, $Q(4, 0)$ and $R(0, -5)$ form a triangle. Find the length of the median from Q .

Worked solution.

Step 1: Find the midpoint of PR .

$$M = (0, -1)$$

A median goes from a vertex to the midpoint of the opposite side.

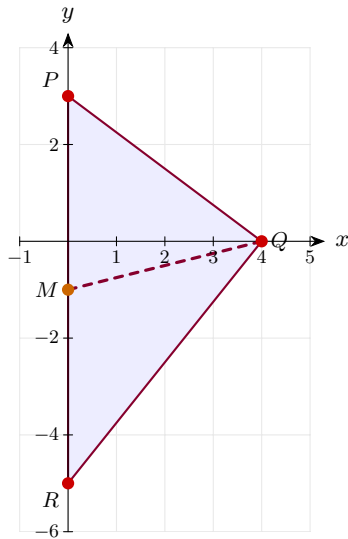
Step 2: Find QM .

$$QM = \sqrt{16 + 1} = \sqrt{17}$$

Distance from $Q(4, 0)$ to $M(0, -1)$.

Final answer: $\sqrt{17}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 56 (cg2-056)

Difficulty: Challenge / Marks: 4

The points $A(1, 3)$ and $B(7, 11)$ are the endpoints of a diameter of a circle. Find the centre and radius.

Worked solution.

Step 1: The centre is the midpoint of the diameter.

$$C = (4, 7)$$

The centre of a circle is at the midpoint of any diameter.

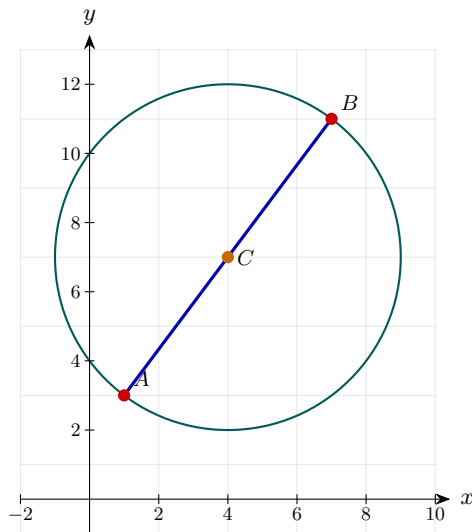
Step 2: The radius is half the diameter length.

$$AB = \sqrt{36 + 64} = 10, \quad r = 5$$

Find the full diameter then halve it.

Final answer: Centre $(4, 7)$, radius 5

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 57 (cg2-057)

Difficulty: Challenge / Marks: 3

The point P divides AB in the ratio $2 : 5$, where $A = (-3, 4)$ and $B = (11, -10)$. Find P .

Worked solution.

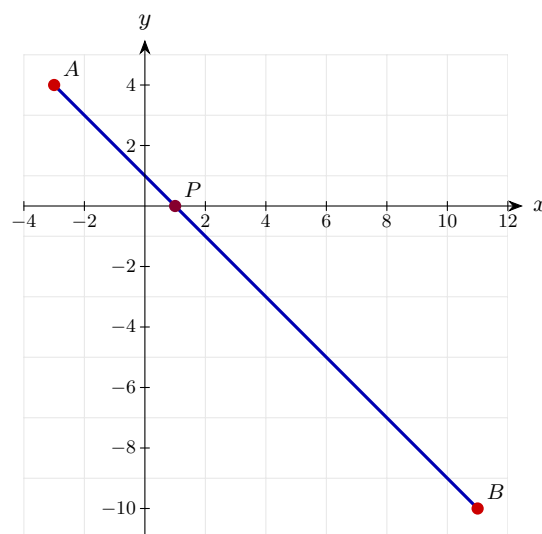
Use the section formula.

$$P = \left(\frac{2(11) + 5(-3)}{7}, \frac{2(-10) + 5(4)}{7} \right) = (1, 0)$$

Section formula for ratio $m : n$: $P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$.

Final answer: $P = (1, 0)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 58 (cg2-058)

Difficulty: Challenge / Marks: 6

A quadrilateral has vertices $A(0,0)$, $B(5,0)$, $C(7,4)$, $D(2,4)$. Show it is a parallelogram and find its diagonal lengths.

Worked solution.

Step 1: Show opposite sides are equal.

$$AB = DC = 5, \quad AD = BC = \sqrt{20}$$

Both pairs of opposite sides equal, so it is a parallelogram.

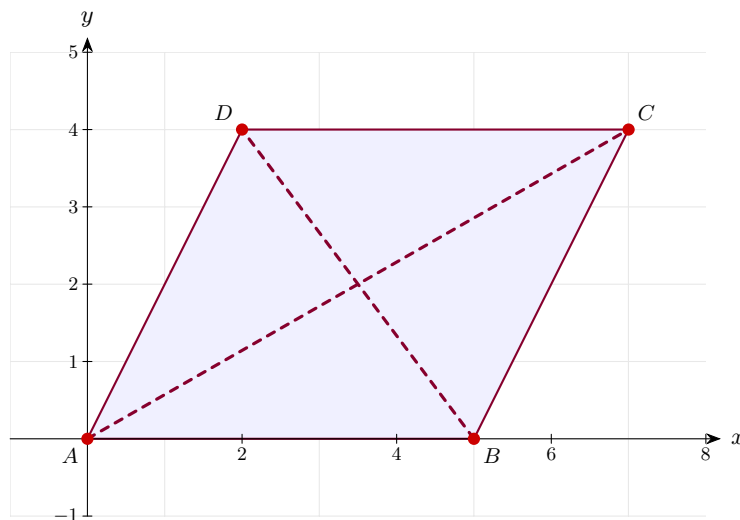
Step 2: Find diagonals.

$$AC = \sqrt{65}, \quad BD = 5$$

Diagonals are not equal, confirming it is not a rectangle.

Final answer: Parallelogram; $AC = \sqrt{65}$, $BD = 5$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 59 (cg2-059)

Difficulty: Challenge / Marks: 5

The points $A(3,1)$ and $B(9,9)$ are joined. C divides AB in ratio $1 : 2$ and D divides AB in ratio $2 : 1$. Find CD .

Worked solution.

Step 1: Find C and D (trisection points).

$$C = \left(5, \frac{11}{3}\right), \quad D = \left(7, \frac{19}{3}\right)$$

C is $\frac{1}{3}$ along, D is $\frac{2}{3}$ along from A to B .

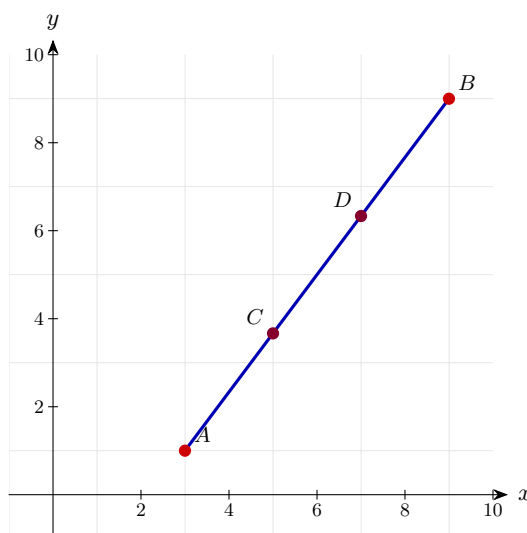
Step 2: Find CD .

$$CD = \sqrt{4 + \frac{64}{9}} = \frac{10}{3}$$

$CD = \frac{1}{3}AB$, which makes sense as C and D are trisection points.

Final answer: $CD = \frac{10}{3}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 60 (cg2-060)

Difficulty: Challenge / Marks: 3

The vertices of triangle PQR are $P(-2, 6)$, $Q(4, 2)$ and $R(0, -4)$. Find the centroid.

Worked solution.

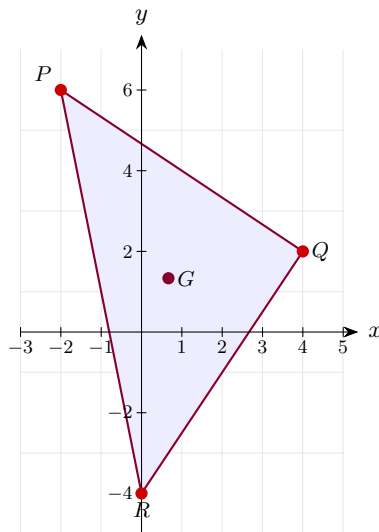
The centroid is the average of the three vertices.

$$G = \left(\frac{-2 + 4 + 0}{3}, \frac{6 + 2 - 4}{3} \right) = \left(\frac{2}{3}, \frac{4}{3} \right)$$

The centroid is where all three medians meet — simply the mean of the coordinates.

Final answer: $G = \left(\frac{2}{3}, \frac{4}{3} \right)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 61 (cg2-061)

Difficulty: Challenge / Marks: 4

A circle has centre $C(3,4)$ and passes through $A(0,0)$. Does $B(6,8)$ lie inside, on, or outside the circle?

Worked solution.

Step 1: Find the radius.

$$r = CA = \sqrt{9 + 16} = 5$$

The radius is the distance from centre to a point on the circle.

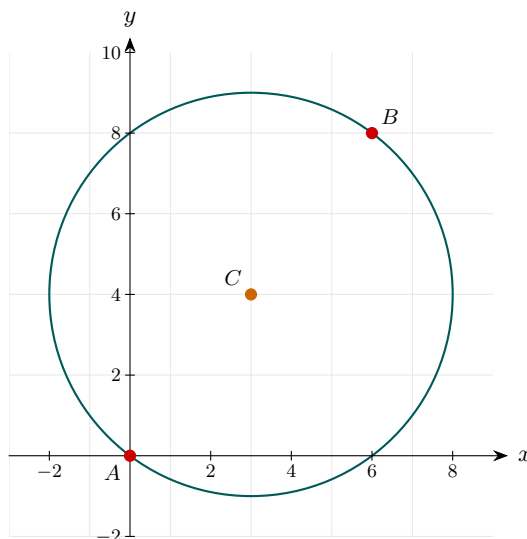
Step 2: Find CB .

$$CB = \sqrt{9 + 16} = 5 = r$$

$CB = r$, so B lies exactly on the circle.

Final answer: B lies on the circle

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 62 (cg2-062)

Difficulty: Challenge / Marks: 5

The points $A(2, -1)$, $B(6, 3)$ and $C(10, -1)$ form a triangle. Find the equation of the perpendicular bisector of AC and verify it passes through B .

Worked solution.

Step 1: AC is horizontal (gradient 0), so the perpendicular bisector is vertical through the midpoint.

$$M_{AC} = (6, -1), \quad \text{perpendicular bisector: } x = 6$$

The perpendicular to a horizontal line is vertical.

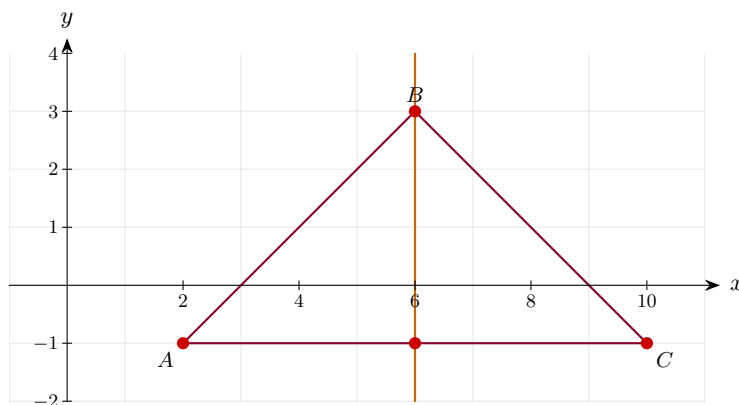
Step 2: Verify $B(6, 3)$ lies on $x = 6$.

B has $x = 6$ ✓

B is equidistant from A and C , confirming the triangle is isosceles.

Final answer: Perpendicular bisector: $x = 6$; B lies on it

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 63 (cg2-063)

Difficulty: Challenge / Marks: 5

AB has $A = (1, 4)$ and $B = (7, -2)$. The point P lies on AB such that $AP = 2$. Find P .

Worked solution.

Step 1: Find the unit direction vector from A to B .

$$AB = 6\sqrt{2}, \quad \hat{u} = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

Direction is $(6, -6)$, length $6\sqrt{2}$, so the unit vector divides by this.

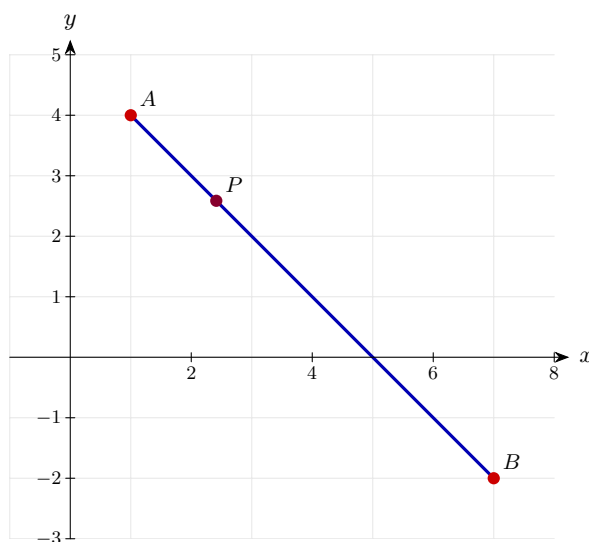
Step 2: Move distance 2 from A.

$$P = (1, 4) + 2\hat{u} = (1 + \sqrt{2}, 4 - \sqrt{2})$$

Multiply unit vector by 2 and add to A.

$$\text{Final answer: } P = (1 + \sqrt{2}, 4 - \sqrt{2})$$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 64 (cg2-064)

Difficulty: Challenge / Marks: 3

Prove that $A(1, 1)$, $B(5, 3)$ and $C(-3, -1)$ are collinear.

Worked solution.

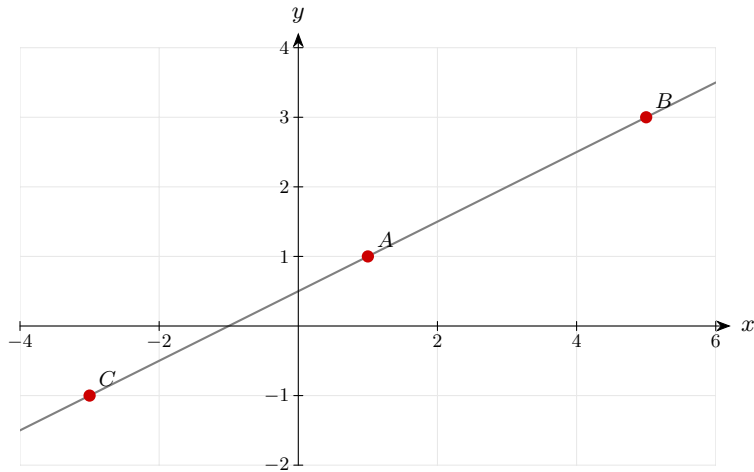
Show the gradients are equal.

$$m_{AB} = \frac{1}{2}, \quad m_{AC} = \frac{1}{2}$$

Two segments sharing a common point with equal gradients means all three points are collinear.

$$\text{Final answer: } m_{AB} = m_{AC} = \frac{1}{2}, \text{ so } A, B, C \text{ are collinear}$$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 65 (cg2-065)

Difficulty: Challenge / Marks: 5

The points $A(0, 6)$, $B(8, 0)$ and $C(3, h)$ form a right angle at C . Find h .

Worked solution.

Step 1: For a right angle at C : $m_{CA} \times m_{CB} = -1$.

$$m_{CA} = \frac{h - 6}{3}, \quad m_{CB} = \frac{-h}{5}$$

Two lines are perpendicular iff the product of their gradients is -1 .

Step 2: Solve the resulting quadratic.

$$\frac{(h - 6)(-h)}{15} = -1$$

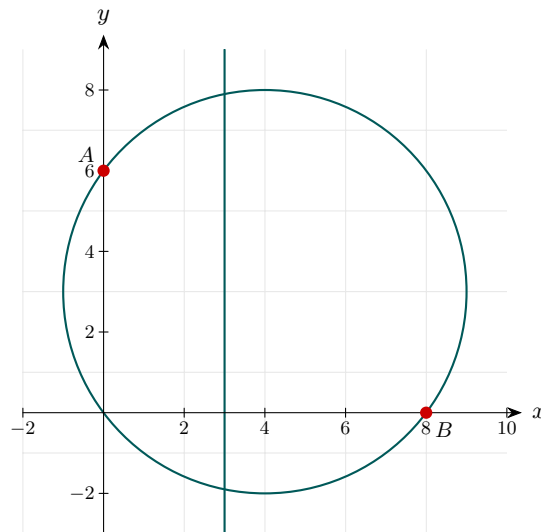
$$h^2 - 6h - 15 = 0$$

$$h = 3 \pm 2\sqrt{6}$$

Both solutions are geometrically valid.

Final answer: $h = 3 + 2\sqrt{6}$ or $h = 3 - 2\sqrt{6}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 66 (cg2-066)

Difficulty: Challenge / Marks: 6

The midpoints of the sides of triangle $A(0, 0)$, $B(8, 0)$, $C(4, 6)$ form a smaller triangle. Find the ratio of its area to the area of ABC .

Worked solution.

Step 1: Find the three midpoints.

$$M_{AB} = (4, 0), \quad M_{BC} = (6, 3), \quad M_{AC} = (2, 3)$$

Apply the midpoint formula to each side of the triangle. These three midpoints form the medial triangle.

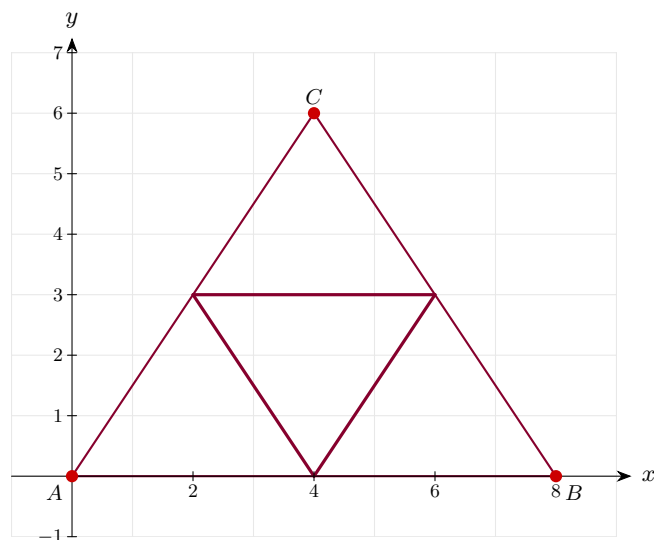
Step 2: Find both areas.

$$\text{Area}_{ABC} = 24, \quad \text{Area}_{\text{medial}} = 6$$

The medial triangle always has $\frac{1}{4}$ the area of the original.

Final answer: $\frac{1}{4}$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 67 (cg2-067)

Difficulty: Challenge / Marks: 4

The segment from $A(2,3)$ to $B(10,7)$ is extended beyond B to C such that $BC = \frac{1}{2}AB$. Find C .

Worked solution.

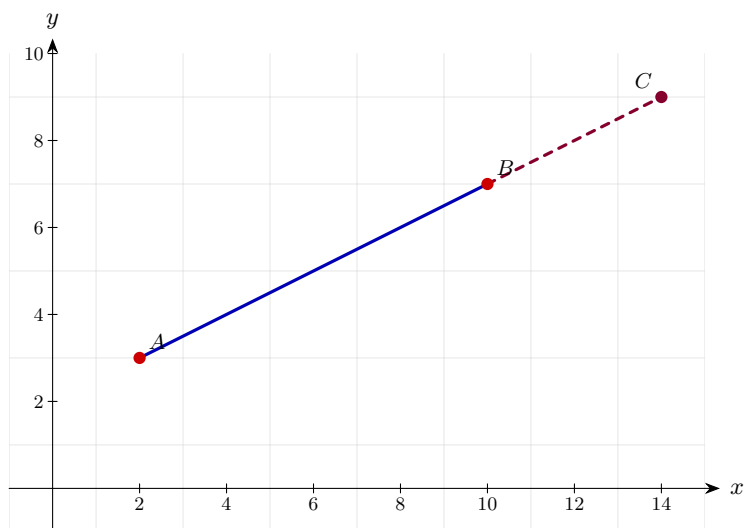
Move half the direction vector beyond B .

$$C = B + \frac{1}{2}(B - A) = (10, 7) + (4, 2) = (14, 9)$$

Since $BC = \frac{1}{2}AB$, add half of $\vec{AB} = (8, 4)$ to B .

Final answer: $C = (14, 9)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 68 (cg2-068)

Difficulty: Challenge / Marks: 6

A rhombus $ABCD$ has $A = (1, 0)$ and $C = (7, 8)$. Given that B lies on the x -axis, find B and D .

Worked solution.

Step 1: Diagonals bisect each other.

$$M = (4, 4)$$

M is also the midpoint of BD .

Step 2: Diagonals are perpendicular.

$$m_{AC} = \frac{4}{3}, \quad m_{BD} = -\frac{3}{4}$$

Negative reciprocals.

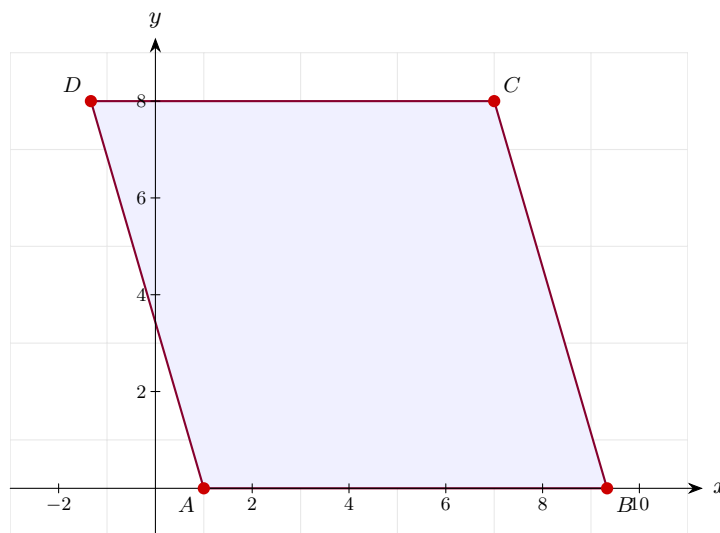
Step 3: B is on $y = 0$ and on the line through M with gradient $-\frac{3}{4}$.

$$x = \frac{28}{3}, \quad D = \left(-\frac{4}{3}, 8\right)$$

Set $y = 0$, solve for x . Then $D = 2M - B$.

Final answer: $B = \left(\frac{28}{3}, 0\right), D = \left(-\frac{4}{3}, 8\right)$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 69 (cg2-069)

Difficulty: Challenge / Marks: 6

Triangle ABC has $A = (0, 0)$, $B = (12, 0)$ and $C = (4, 8)$. Verify that the centroid divides each median in the ratio $2 : 1$.

Worked solution.

Step 1: Find the centroid.

$$G = \left(\frac{16}{3}, \frac{8}{3} \right)$$

Average of the three vertices.

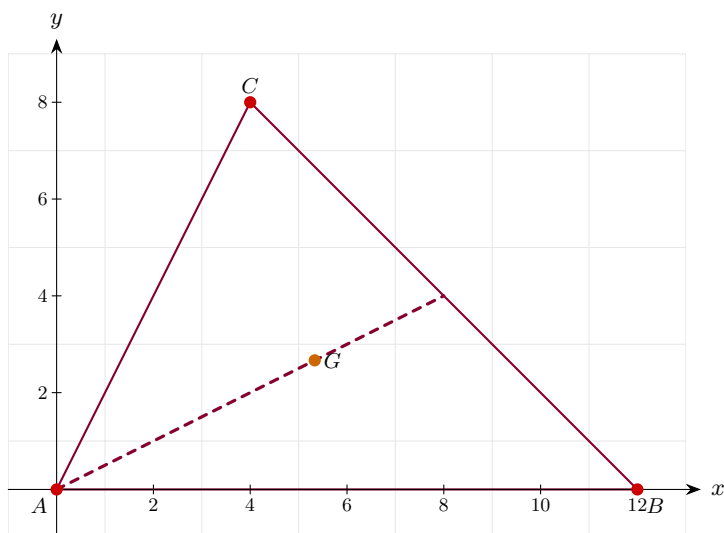
Step 2: Check median from A to midpoint of BC.

$$M_A = (8, 4), \quad AG = \frac{8\sqrt{5}}{3}, \quad GM_A = \frac{4\sqrt{5}}{3}$$

$\frac{AG}{GM_A} = 2$, confirming the 2 : 1 ratio. This is a general property of all triangles.

Final answer: Verified: $AG : GM_A = 2 : 1$

Diagram.



The points and segments shown on the coordinate plane.

Line Segments 70 (cg2-070)

Difficulty: Challenge / Marks: 3

The line segment from $(\cos \theta, \sin \theta)$ to $(-\cos \theta, -\sin \theta)$ has length L . Find L , giving your answer as a single integer.

Worked solution.

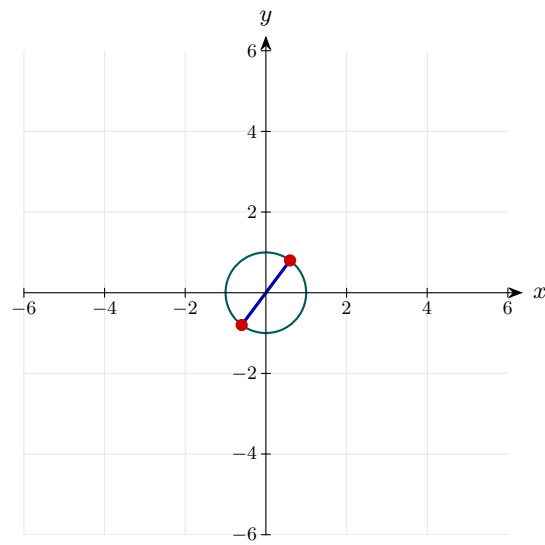
Apply the distance formula and use the Pythagorean identity.

$$\begin{aligned} L &= \sqrt{(2 \cos \theta)^2 + (2 \sin \theta)^2} \\ &= 2\sqrt{\cos^2 \theta + \sin^2 \theta} = 2 \end{aligned}$$

The identity $\cos^2 \theta + \sin^2 \theta = 1$ simplifies the expression to exactly 2, regardless of θ . The two points are diametrically opposite on the unit circle.

Final answer: 2

Diagram.



The points and segments shown on the coordinate plane.
